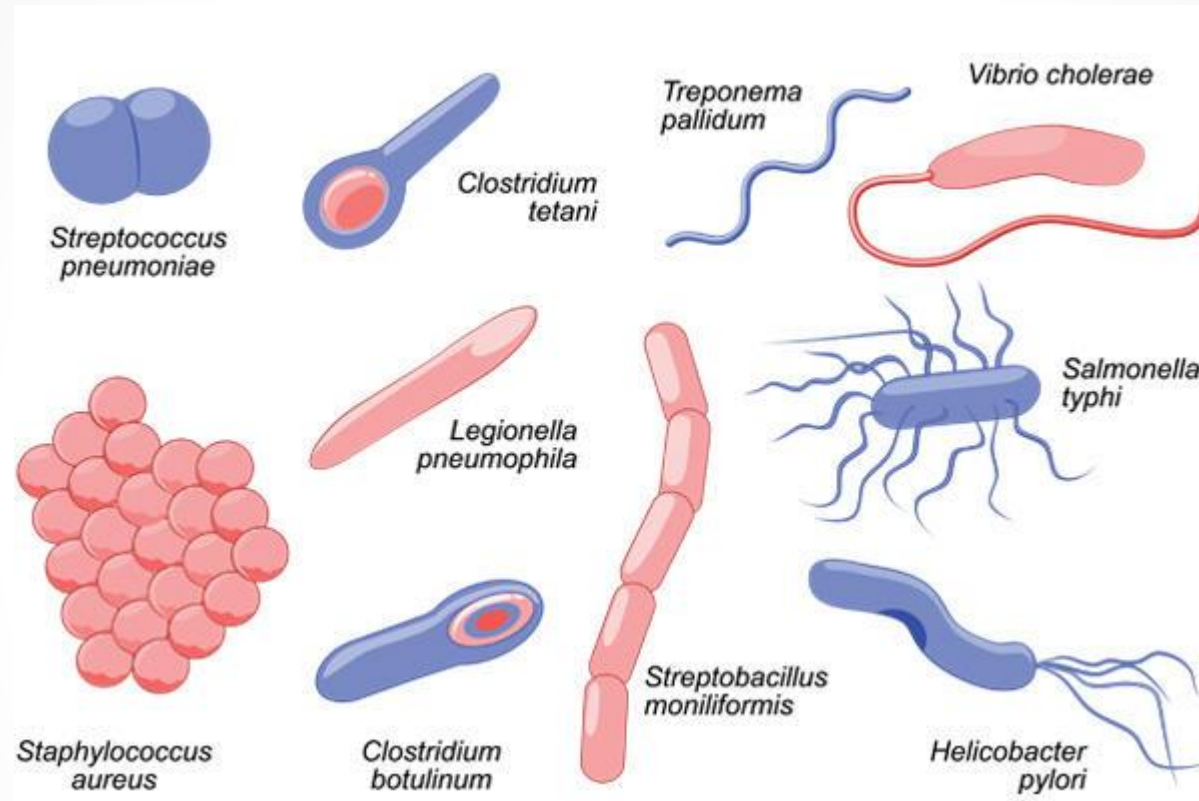


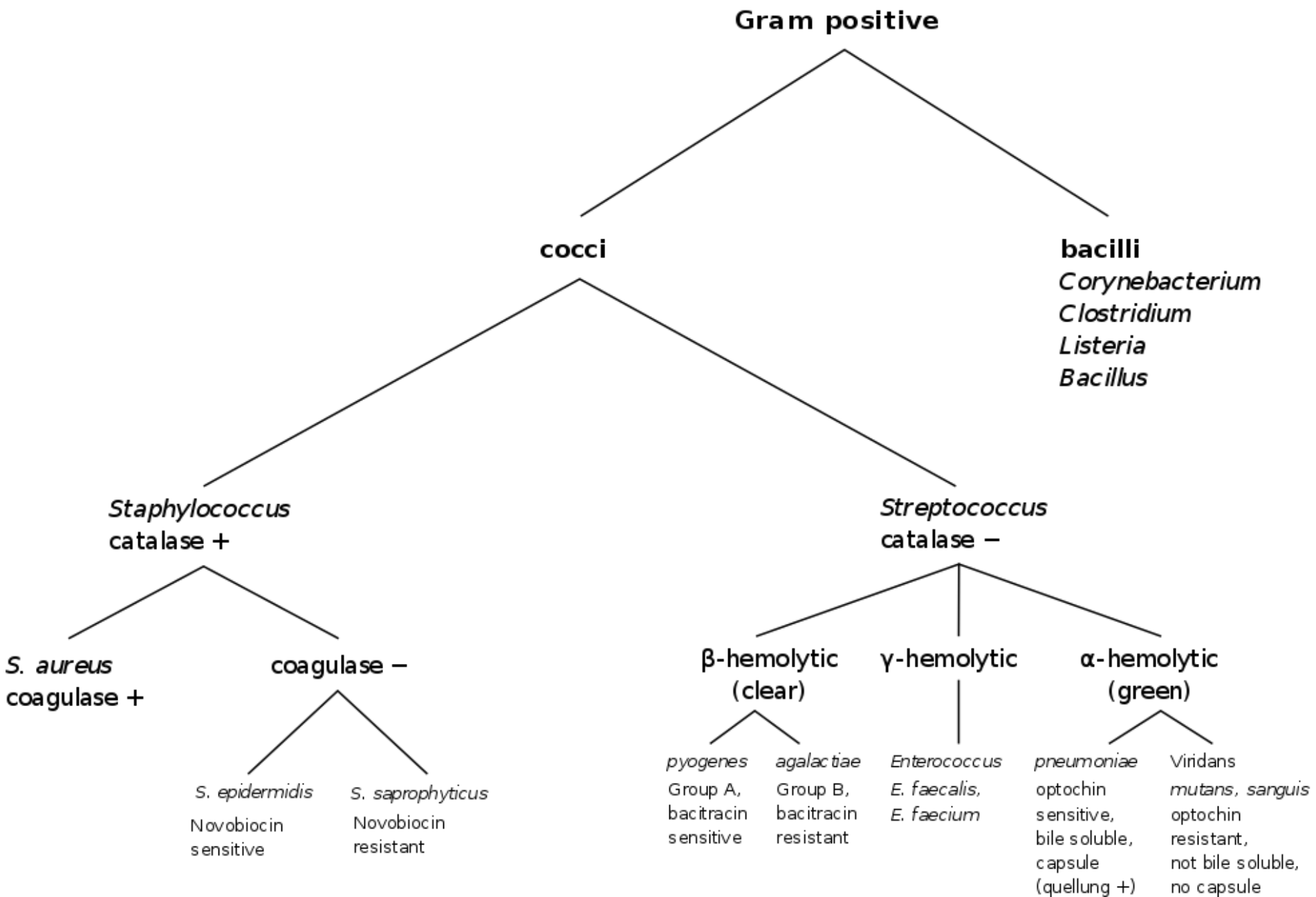
Clinical Bacteriology



Dr. Alem A (PhD, Medical Microbiology)

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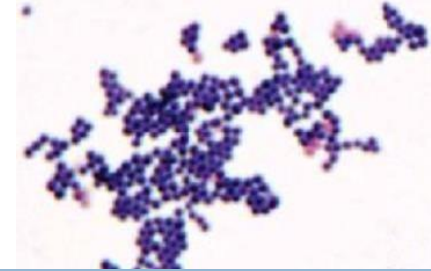
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Staphylococci

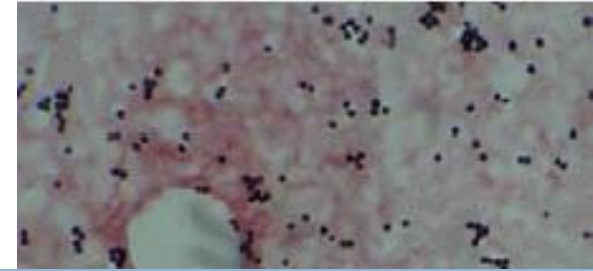
Staphylococci



9

- Gram positive, spherical cocci that occur in grape-like clusters
- Nonmotile, nonsporeforming, facultative anaerobic
- Are **catalase** positive
- Grow readily on many types of culture media
- There are more than 50 species and subspecies of staphylococci
 - _ ubiquitous in the air, dust, sewage, water, environmental surfaces, humans and animals
 - _ classified in to **coagulase positive** staphylococci (CoPS) and **Coagulase negative** staphylococci (CoNS) based on their **ability to produce coagulase**

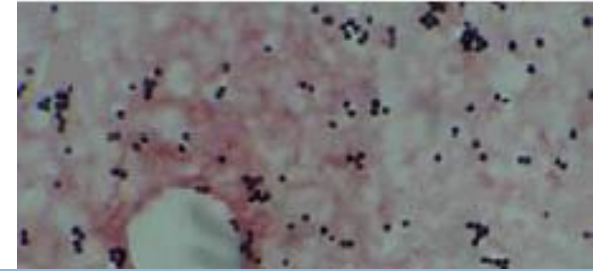
Staphylococcus aureus



12

- usually colonizes the skin and mucosa of human and warm-blooded animals → Commensal
- a third of world population is colonized by *S. aureus*
 - _ the most common carriage site is in the anterior nares
 - _ mucosal surfaces: throat, vaginal wall, & Gastrointestinal tract
 - _ warm and moist skin folds: such as behind ears, under pendulous breasts, and in the groin
- Animals: anterior nares, mucosal surfaces and udders
- Dairy animals shed staphylococci in milk, permanently or intermittently

S. aureus...



- invasion by *S. aureus* usually results from
 - epithelial lining injury, trauma, medical or surgical interventions, as well as viral infections
 - Food poisoning results from ingestion of heat stable staphylococcal enterotoxins along with food
- *S. aureus* possess a varying repertoire of virulence factors, many of which are encoded on MGEs
 - Cell surface factors
 - Toxins
 - Secreted factors/enzymes

Pathology depends on the ability of the bacteria to

- evade phagocytosis
- produce surface proteins that mediate adherence of the bacteria to host tissues
- elaboration of specific toxins and hydrolytic enzymes

Staphylococcus aureus

Virulence factors

- I. Capsule: inhibit phagocytosis
- II. Surface protein A: Binds to the Fc portion of IgG thus effectively diminishing opsonization
- III. Cytotoxins: binds to the membrane of susceptible cells resulting in lysis
- IV. Coagulase: convert fibrinogen to fibrin clot → retard migration of phagocytes to the site of infection
- V. Exfoliative toxins: causes intercellular splitting of the epidermis by disrupting intercellular junctions
- VI. Superantigens toxins: stimulates both T cells and macrophages to release massive amounts of cytokines → shock

Diseases caused by *S. aureus*

8

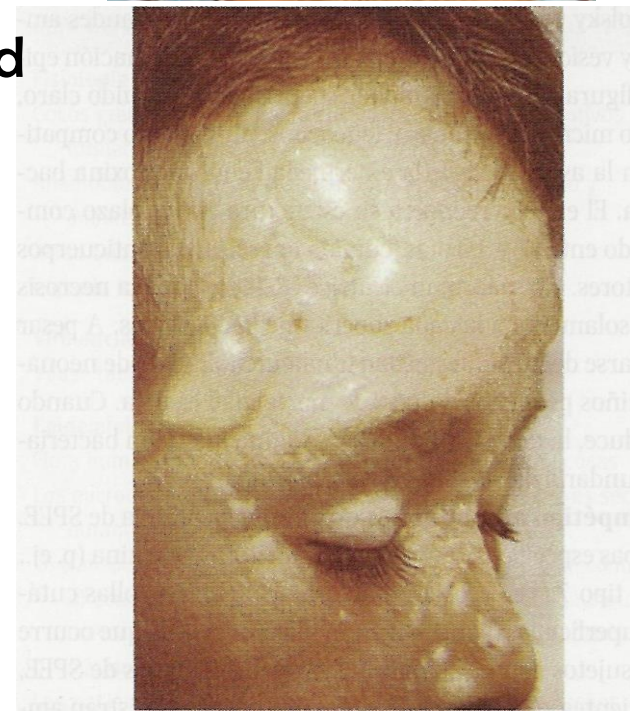
a. Impetigo

- localized cutaneous infection primarily on the face and limbs characterized by pus-filled vesicle on an erythematous base
- Crusting occurs after the pustule ruptures resulting in the characteristic honey-crusted exudate



b. Bullous impetigo

- characterized by large blisters containing many staphylococci in the superficial layers of the skin
- Bullous impetigo can be considered as a localized form of scalded skin syndrome



Diseases caused by *S. aureus*...

c. Folliculitis

- a pyogenic infection of the hair follicles
- manifest as clusters of small erythematous papules or pustules
- Inflammation is superficial and pus is only present in the epidermis

d. Furuncles(boils)

- an extension of folliculitis
- large, painful, raised nodules that have an underlying collection of dead and necrotic tissue

e. Carbuncles

- occur when furuncles coalesce and extend to the deeper subcutaneous tissue
- characterized by multiple drainage points
- Unlike patients with folliculitis and furuncles, patients with carbuncles have **chills and fevers**

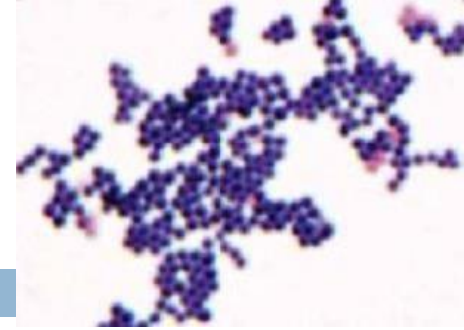


Diseases caused by *S. aureus*...

- f. Bacteremia: Presence of bacteria in the blood
- g. Deep Lesions
 - *S. aureus* can cause a wide variety of infections of deep tissues by **bacteremic spread** from a skin lesion → bones, joints, deep organs, and soft tissues
- h. Scalded Skin Syndrome: due to exfoliative toxin
- i. Staphylococcal Toxic Shock Syndrome (STSS)
 - Due to *tsst-1* and enterotoxins
- j. Staphylococcal food poisoning:
 - due to enterotoxins



Laboratory diagnosis of *S. aureus*



- **Microscopy:** gram +ve cocci in cluster
- **Culture:** from skin lesion, Aspirates and blood cultures on blood agar

Treatment

- Superficial lesions resolve spontaneously
- Penicillinase-resistant β -lactams (methicillin, oxacillin) used pending susceptibility tests
- Vancomycin is often used to treat serious infections with methicillin resistant *S. aureus* (MRSA)

Coagulase negative staphylococci (CoNS)

- Most Common species: *S. epidermidis*, *S. saprophyticus*, *S. lugdunensis*, *S. haemolyticus*
- Normal human flora on skin and mucosal surfaces
- Can survive on dry surfaces for long periods
- Relatively avirulent
- production of a “slime layer” can allow adherence to foreign bodies (e.g., catheters, grafts, prosthetic valves and joints, shunts)
- Patients are at risk when a foreign body is present

CNS associated endocarditis

- The CNS can infect native and prosthetic heart valves
 - *S. lugdunensis* is most frequently associated with native valve endocarditis
- The CNSs are major cause of endocarditis of artificial valves
 - The organisms are introduced at the time of valve replacement
 - Infection occurs more commonly at the site where the valve is sewn to the heart tissue
 - Infection with abscess formation can lead to separation of the valve at the suture line → mechanical heart failure
 - *S. epidermidis*, *S. lugdunensis*, *S. haemolyticus*

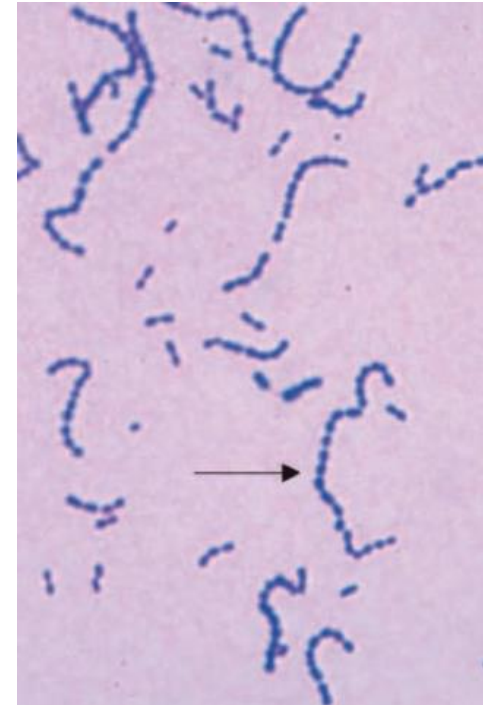
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Streptococci

Streptococci

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- Diverse collection of spherical gram-positive cocci arranged in chains or pairs
- All are catalase negative unlike staphylococci
- facultative anaerobes
- Ferment carbohydrates
- Have complex nutritional requirements → necessitates the use of blood- or serum-enriched media for isolation
- more than 100 species within the genus *Streptococcus*



Streptococci

16

Classification

- **complicated because three different overlapping schemes are used:**
 1. **Serologic properties: Lancefield groupings (originally A to W);**
 2. **Hemolytic patterns (α , β , γ)**
 3. **Biochemical (physiologic) properties**

Streptococci

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Lancefield groupings

- Developed by Rebecca Lancefield developed in 1933
- Serologic classification
- Most Streptococci possess group-specific cell wall antigens, most of which are carbohydrates
 - These antigens can be readily detected by immunologic assays

Group	Representative Species	Diseases
A	<i>S. pyogenes</i>	Pharyngitis, skin and soft-tissue infections, bacteremia, rheumatic fever, acute glomerulonephritis
	<i>S. anginosus</i> group	Abscesses
B	<i>S. agalactiae</i>	Neonatal disease, endometritis, wound infections, urinary tract infections, bacteremia, pneumonia, skin and soft-tissue infections
C	<i>S. dysgalactiae</i>	Pharyngitis, acute glomerulonephritis
F, G	<i>S. anginosus</i> group	Abscesses
	<i>S. dysgalactiae</i>	Pharyngitis, acute glomerulonephritis

Streptococci

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Hemolytic patterns

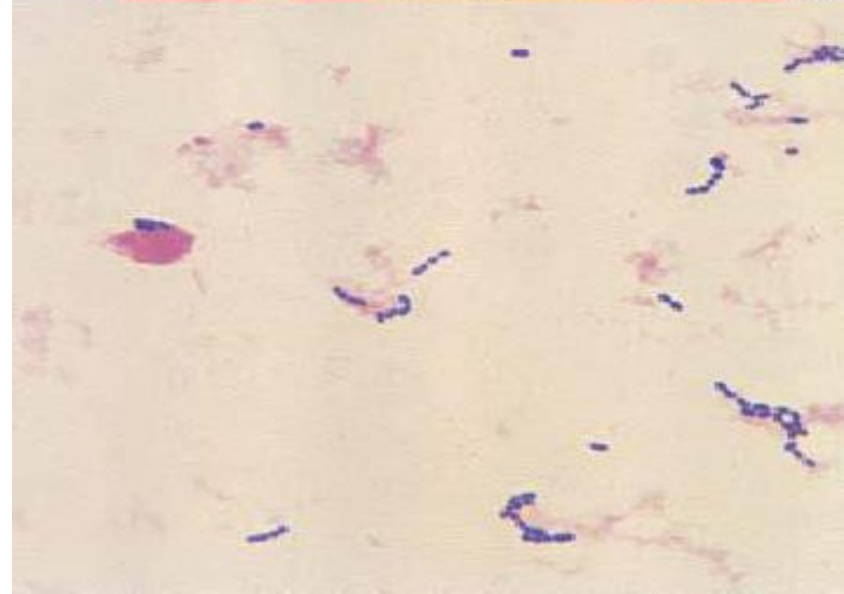
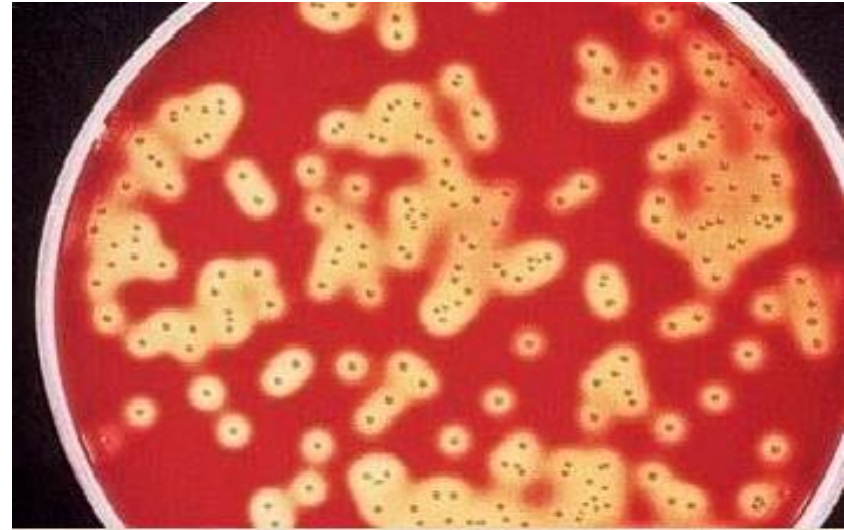
- Many streptococci are able to hemolyze RBCs in vitro in varying degrees
- A. β -Hemolytic streptococci
 - complete lysis of RBCs, i.e. form a clear zone around their colonies
 - β -Hemolysis is due to the production of enzymes (hemolysins) called streptolysin O and streptolysin S
- B. α -Hemolytic streptococci
 - incomplete lysis of red blood cells
 - form a green zone around their colonies $\rightarrow$ H_2O_2 produced by the bacteria oxidizes hemoglobin (red color) to biliverdin (green color)
- C. γ -hemolysis
 - No lysis of RBCs (Nonhemolytic)

Streptococcus pyogenes

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General characteristics

- Gram +ve cocci arranged in chains
- Grow best in enriched media under aerobic or anaerobic Conditions (facultative)
- Form β -hemolytic colonies on blood agar



Virulence factors *S. pyogenes*

- a. Capsule: interferes with phagocytosis
- b. M protein: binds to the serum factor H → prevent phagocytosis
- c. C5a peptidase: inactivates C5a which is a chemo-attractant of neutrophils and mononuclear phagocytes
- d. Pyrogenic exotoxins: act as Superantigens
- e. Streptolysin (O and S): lyse erythrocytes, leukocytes, and platelets

Diseases caused by *S. pyogenes*

a. Pharyngitis

- generally develops 2 to 4 days after exposure
- abrupt onset of sore throat, fever, malaise, and headache
- posterior pharynx appear erythematous with an exudate



b. Pyoderma (impetigo)

- Like that of impetigo caused by *S. aureus*

d. Erysipelas

- painful, well-demarcated, superficial infection of the dermis that involves the cutaneous



Streptococcal Disease...

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e. Cellulitis

- an acute infection involving primarily the dermis and subcutaneous tissues
- Lesions are characterized by edema, pain, tenderness, warmth, erythema, and spreading with irregular margins

e. Necrotizing fasciitis (streptococcal gangrene)

- occurs deep in the subcutaneous tissue
- characterized by an extensive destruction of muscle and fat
- Can be also caused by *S. aureus*
- Is a medical emergency requiring prompt surgical debridement



Streptococcal Disease...

23

h. Toxic shock syndrome

i. Bacteremia

- *S. pyogenes* is commonly isolated in blood cultures
- Common in most patients with necrotizing fasciitis or toxic shock syndrome
- mortality in this population of patients approaches 40%

Non-suppurative Streptococcal Disease

a. Rheumatic fever

- an autoimmune, complication of *S. pyogenes* pharyngitis
- Affects heart, joints, blood vessels
- Antistreptococcal antibodies cross-react with heart sarcolemma = M protein and myosin structural similarity

b. Acute glomerulonephritis

- acute inflammation of the renal glomeruli with edema, hypertension, hematuria, and proteinuria
- caused by deposition of antigen–antibody complexes with complement activation and consequent inflammation in the glomerulus

Laboratory Diagnosis of *S. pyogenes*

Microscopy: gram-positive cocci in pairs and chains in association with leukocytes

Culture

- *S. pyogenes* is susceptible to bacitracin
- β -hemolytic

Antibody Detection

- ASO test: measuring antibodies against streptolysin O for confirming rheumatic fever or acute glomerulonephritis

Treatment: Penicillin, cephalosporins



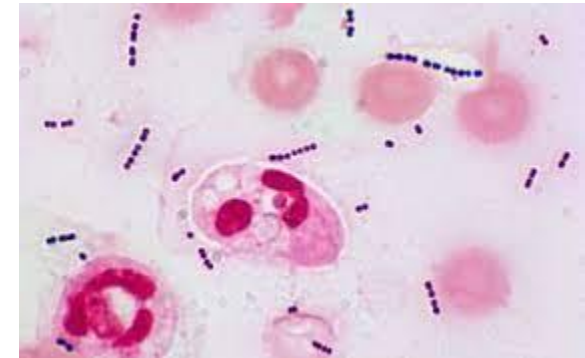
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***Streptococcus agalactiae* (Group B streptococci)**

Streptococcus agalactiae

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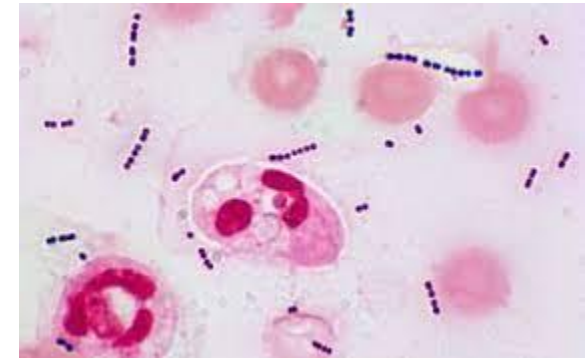
- the only species that has the group B antigen
- gram-positive cocci (0.6 to 1.2 μm) that form short chains in clinical specimens and longer chains in culture
 - make it indistinguishable on Gram stain from *S. pyogenes*
- grow well on nutritionally enriched media
- in contrast with the colonies of *S. pyogenes*, the colonies of *S. agalactiae* are large with a narrow zone of β -hemolysis



Streptococcus agalactiae

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- Strains of *S. agalactiae* can be characterized on the basis of three serologic markers
 1. Group-specific cell wall polysaccharide B antigen (Lancefield grouping antigen)
 2. Nine type-specific capsular polysaccharides (Ia, Ib, and II to VIII)
 3. Surface proteins (the most common is the c antigen)



S. agalactiae...

29

Pathogenesis and Immunity

A. Polysaccharide capsule

- interferes with phagocytosis until the patient develops type-specific antibodies
- Antibodies against the type-specific capsular antigens are protective
- partly explains the predilection of this organism for neonates
- In the absence of maternal antibodies, the neonate is at risk for disease
- Genital colonization with GBS has been associated with increased risk of premature delivery
- Premature infants are at greater risk of disease due to GBS

S. agalactiae...

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Epidemiology

- GBS colonizes the lower gastrointestinal tract and the genitourinary tract
- Transient vaginal carriage observed in 10-30% of pregnant women
- Approximately 60% of infants born to colonized mothers become colonized with their mothers' organisms
- Risk factors for infant colonization
 - Mothers colonized with large numbers of bacteria
 - Premature delivery
 - Prolonged membrane rupture
 - intrapartum fever

S. agalactiae...

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Epidemiology

- Colonization with subsequent development of disease in the neonate can occur
 - in utero, at birth, or during the first few months of life
- *S. agalactiae* is the most common cause of bacterial septicemia and meningitis in newborns
- The risk of invasive disease in adults is greater in pregnant women than in men and nonpregnant women

S. agalactiae...

32

Clinical Diseases

- Early-Onset Neonatal Disease
 - GBS acquired in utero or at birth
 - Develop during the first week of life
 - characterized by bacteremia, pneumonia, or **meningitis**
- Late-Onset Neonatal Disease
 - acquired from an exogenous source (e.g., mother, another infant) and develops between 1 week and 3 months of age
 - Predominant manifestation is **bacteremia with meningitis**

S. agalactiae...

33

Clinical Diseases

- **Infections in Pregnant Women**

- Postpartum endometritis, wound infection, and urinary tract infections

- **Infections in Men and Non-pregnant Women**

- Compared with pregnant women who acquire GBS infection, men and non-pregnant women with GBS infections are generally **older and have debilitating underlying conditions**
- The most common presentations are bacteremia, pneumonia, bone and joint infections, and skin and soft-tissue infections
- Because these patients often have compromised immunity, mortality is higher in this population

S. agalactiae...

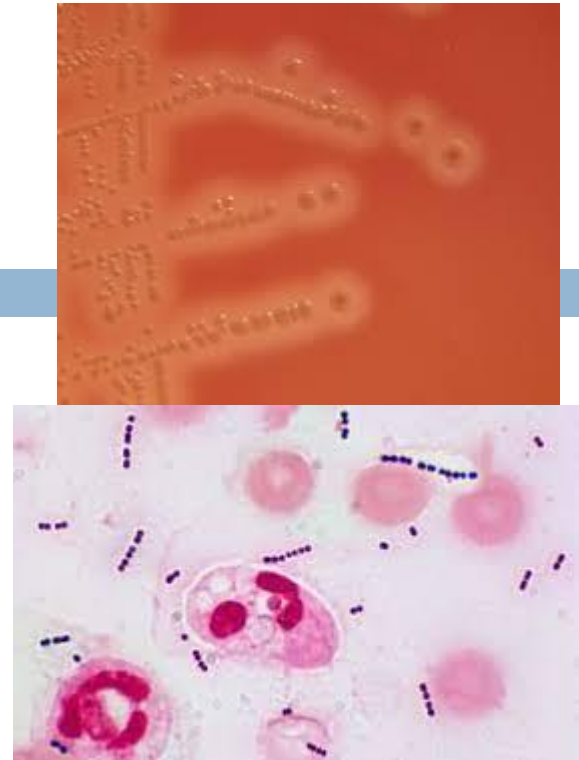
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Laboratory Diagnosis

- Culture: vaginal, rectal swab
- PCR

Treatment and prevention

- Penicillin
- Cephalosporin or vancomycin can be used in penicillin-allergic patients
- In an effort to prevent neonatal disease, it is recommended that all pregnant women should be screened for colonization with GBS at 35 to 37 weeks' gestation (CDC)
- Chemoprophylaxis should be used for all women who are either colonized or at high risk
- Intravenous penicillin G or ampicillin administered at least 4 hours before delivery is recommended



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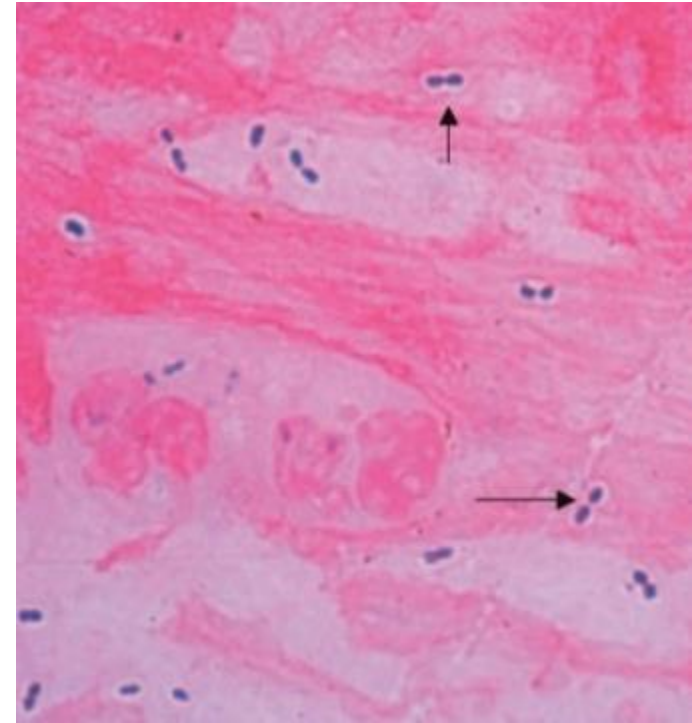
Streptococcus pneumoniae

- Pneumonia

Streptococcus pneumoniae (Pneumococci)

36

- Pneumococci are gram-positive lancet-shaped cocci arranged in pairs (diplococci) or short chains
 - oval with somewhat pointed ends rather than being round
- On blood agar, they produce α -hemolysis
- In contrast to viridans streptococci, they are lysed by bile or deoxycholate, and their growth is inhibited by optochin
- >94 serotypes



Streptococcus pneumoniae...

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Virulence factors

- **Polysaccharide capsules**
 - interfere with phagocytosis and favor invasiveness
 - elicits primarily a β -cell (i.e., T-independent) response
 - Specific antibody to the capsule opsonizes the organism, facilitates phagocytosis
- **Teichoic acid:** activates complement and induces inflammatory cytokine production
- **Pneumolysin:** a cytotoxin similar to the streptolysin O in *S. pyogenes*, binds cholesterol in the host cell membrane and creates pores
- **IgA protease:** enhances the organism's ability to colonize the mucosa of the upper respiratory tract by cleaving IgA

Streptococcus pneumoniae...

38

Transmission

- Humans are the natural hosts for pneumococci → there is no animal reservoir
- common inhabitant of the throat and nasopharynx in healthy people, with colonization more common in children than in adults
 - not considered to be communicable
- Pneumococcal disease occurs when organisms colonizing the nasopharynx and oropharynx spread to
 - the lungs (pneumonia)
 - paranasal sinuses (sinusitis)
 - ears (otitis media)
 - meninges (meningitis)

Streptococcus pneumoniae...

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Clinical Diseases

- **Pneumonia**
 - often begins with a sudden chill, fever, cough, and pleuritic pain
 - Sputum is a red or brown “rusty” color.
- **Sinusitis and Otitis Media**
- **Bacteremia**
 - occurs in 25% to 30% of patients with pneumococcal pneumonia and in more than 80% of patients with meningitis
- **Meningitis**

Laboratory Diagnosis

Microscopy

- Elongated pairs of gram-positive cocci surrounded by an unstained capsule
- Quellung (German for “swelling”) reaction
 - polyvalent anticapsular antibodies are mixed with the bacteria
 - the mixture is examined microscopically
 - A greater refractiveness around the bacteria is a positive reaction for *S. pneumoniae*
- An alternative test
 - Mix a drop of bile with a suspension of bacteria
 - Bile will dissolve *S. pneumoniae* and no organisms will be seen in the Gram stain

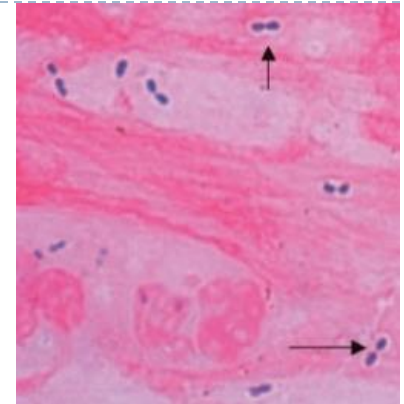


Figure 1 Gram stain of a sputum sample showing *Streptococcus pneumoniae* as gram-positive diplococci.



Clinical Infectious Diseases, Volume 45, Issue 4, 15 March 2008, Pages 555-557, <https://doi.org/10.1093/cid/cin326>
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Laboratory Diagnosis

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- Culture
 - Specimen: Sputum, Blood, CSF etc
 - α -hemolytic colonies on blood agar
 - The colonies are bile-soluble (i.e., are lysed by bile)
 - inhibited by optochin
- PCR



Treatment and prevention

42

Treatment

- Drug of choice for severe pneumococcal infections is penicillin G, for mild infections oral penicillin V
- Erythromycin or one of its long-acting derivatives (e.g., azithromycin) can be used for Penicillin-allergic patients
- Vancomycin is the drug of choice for the penicillin-resistant pneumococci

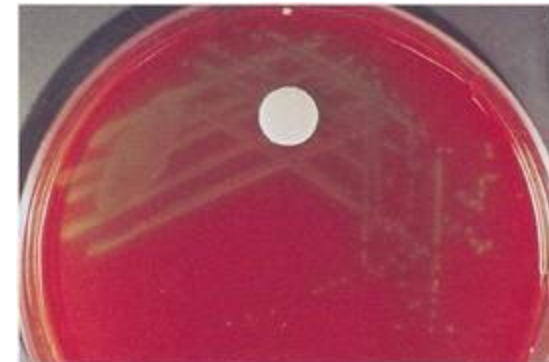
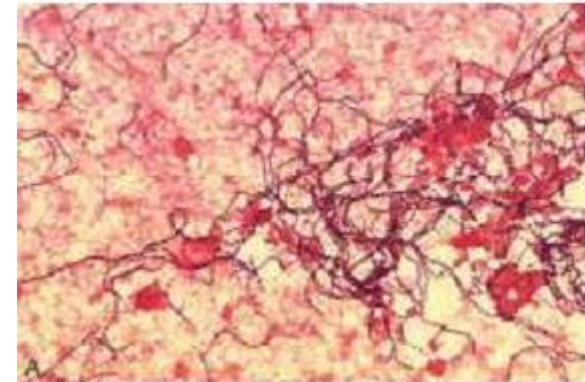
Prevention

- 13-valent pneumococcal conjugate vaccine (Pneumovax 13)
- Unconjugated 23-valent pneumococcal vaccine (Pneumovax 23)

Viridans Streptococci

43

- Heterogeneous collection of streptococci
 - Gram positive cocci arranged in chains
 - α /non-hemolytic
 - uninhibited by optochin
 - not bile soluble
- The group name Viridans is derived from viridis (Latin for "green")
 - many of these bacteria produce a green pigment on blood agar $\rightarrow$ α -hemolysis



Viridans Streptococci

44

- More than 30 species and subspecies have been identified
- colonize the oropharynx, gastrointestinal tract, and genitourinary tract → commensals
 - rarely found on skin surface → surface fatty acids are toxic to them
- are nutritionally fastidious
- require an incubation at atmosphere of 5-10% CO₂

Viridans Streptococci....

45

- Most common species are *S. salivarius*, *S. mutans*, *S. mitis*, and *S. sanguis*
- On access to the blood stream, these organisms are the most frequent cause of sub-acute endocarditis
 - *S. salivarius*
 - *S. mitis*
- How do the Viridans get access to blood stream?
 - Trauma: during dental procedures, vigorous teeth cleaning, flossing

Enterococchi

Enterococci

- were previously classified as group D streptococci
- Gram-positive cocci arranged in pairs and short chains
- Facultative anaerobic that grow in a broad temperature range (10-45° C)
- Form large colonies that appear nonhemolytic, α -hemolytic, or rarely β -hemolytic on blood agar
- can grow in high concentrations of NaCl and bile salts
- can be differentiated by bacitracin resistance, and growth in 40% bile, pH 9.6 or 6.5% salt solution

Enterococci...

- Reclassified into the new genus *Enterococcus* in 1984
- Currently, there are 38 species in this genus
- Part of the normal gastrointestinal and genitourinary tract flora in humans and animals
- Relatively few species are important human pathogens
- The most commonly isolated and clinically important species
 - *Enterococcus faecalis*
 - *Enterococcus faecium*
 - *Enterococcus gallinarum*
 - *Enterococcus casseliflavus*

Enterococci...

- Most infections emerge from patient's bacterial flora
- Individuals at increased risk of enterococci infection include
 - hospitalized for prolonged periods
 - treated with broad-spectrum antibiotics (particularly cephalosporins, to which enterococci are naturally resistant)
- Diseases include
 - urinary tract infections
 - wound infections (particularly intra-abdominal)
 - bacteremia → endocarditis
- Common nosocomial cause of sub-acute endocarditis

Gram-positive spore-forming bacilli

- **Bacillus**
- **Clostridium**

Gram-positive spore-forming bacilli

Introduction

- The Gram-positive spore-forming bacilli consists the *Bacillus* and *Clostridium* species
- They are ubiquitous (present every where)
- Because they form spores they can survive in the environment for many years
- *Bacillus* species are aerobes, clostridia are anaerobes
- Medically important *Bacillus*: *B. anthracis* and *B. cereus*
- *Clostridium*: *C. perfringens*, *C. tetani*, *C. butulinum*, *C. difficile*

Bacillus anthracis

Virulence factors

- A. Capsule: inhibit phagocytosis
- B. Edema toxin: responsible for the fluid accumulation (edema) observed in anthrax by increasing the intracellular cAMP
- C. Lethal toxin: mediates cell lysis
 - Anthrax is primarily a disease of herbivores i.e. goats, sheep, cattle, horses
 - Soil is contaminated with anthrax spores from the carcasses of dead animals
 - Humans become infected incidentally by contact with infected animals or their products

Pathogenesis: *B. anthracis*

- Human anthrax is acquired by one of the following three routes:
 - Inoculation → cutaneous anthrax
 - Ingestion → Gastrointestinal anthrax
 - inhalation of spores: inhalation anthrax
- The spores germinate in the tissue at the site of entry
- growth of the vegetative organisms results in formation of a gelatinous edema and congestion
- Bacilli spread via lymphatics to the bloodstream



Laboratory Diagnosis

Microscopy

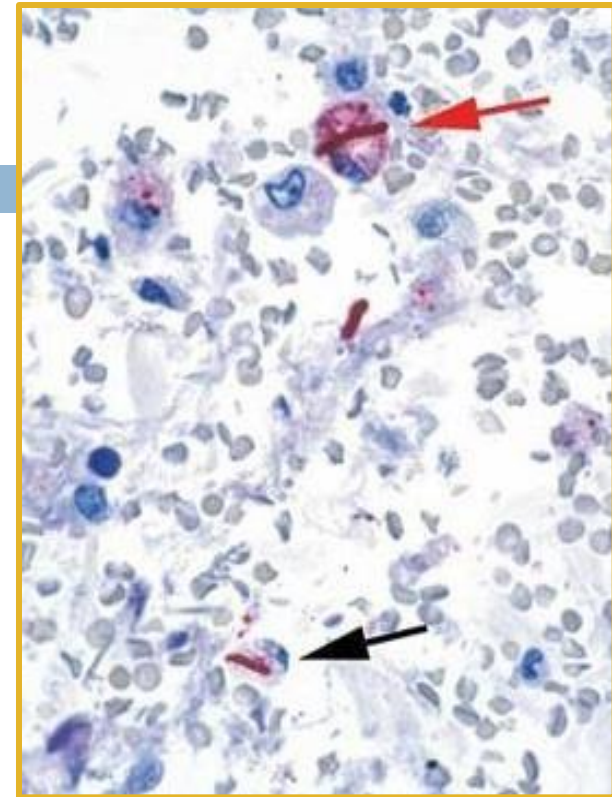
- long, thin, gram-positive rods arranged singly or in long chains
- The capsule can be observed with a contrasting stain such as India ink

Culture

- non hemolytic colonies on blood agar

Treatment

- Penicillin, Doxycycline, and Ciprofloxacin



Clostridium perfringens

Virulence factors

I. α -toxin

- lyses erythrocytes, platelets, leukocytes, and endothelial cells

II. Enterotoxin → Food poisoning

- produced during the phase transition from vegetative cells to spores
- released with the formed spores
- The alkaline conditions in the small intestine stimulate sporulation

C. perfringens: Clinical Diseases

Clostridial myonecrosis (gas gangrene)

- Generally develops within a week after clostridia are introduced into tissue by trauma or surgery
- The onset is followed rapidly by extensive muscle necrosis, shock, renal failure, and death, often within 2 days of initial onset
- production of large amounts of hydrogen and carbon dioxide gas (Gas gangrene)



C. perfringens...

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Food poisoning

- is the second disease caused by *C. perfringens*
- Spores are located in soil and can contaminate food
- The heat-resistant spores survive cooking and germinate
- The organisms grow to large numbers in reheated foods, especially meat dishes
- *C. perfringens* is a member of the normal flora in the colon but not in the small bowel, where the enterotoxin acts to cause diarrhea
- The mode of action of the enterotoxin is the same as that of the enterotoxin of *S. aureus* (i.e., it acts as a superantigen)

C. perfringens...

58

Laboratory Diagnosis

- Microscopy:
 - Large gram-positive rods in smears of tissue and exudate samples
- Culture
 - Cultured anaerobically

Treatment

- Penicillin G is the antibiotic of choice
- Wounds should be debrided
- Food poisoning resolves by itself in 24 hours

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Clostridium tetani

Clostridium tetani

60

Properties

- large, motile, spore-forming rod
- produces round, terminal spores that give it the appearance of a drumstick
- difficult to grow because the organism is extremely sensitive to oxygen toxicity
 - the organisms sporulate readily and can survive in nature for a long time
- *C. tetani* is ubiquitous; it is found in fertile soil



Clostridium tetani

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Pathogenesis and Immunity

- *C. tetani* produces two toxins
 - Tetanolysin: oxygen-labile hemolysin
 - Tetanospasmin: plasmid-encoded, heat-labile neurotoxin
- Tetanospasmin (Tetanus toxin)
 - produced by vegetative cells at the wound site
 - carried intra-axonally (retrograde) to the CNS, where it binds to ganglioside receptors and blocks release of inhibitory mediators (e.g., glycine and γ -aminobutyric acid [GABA]) at spinal synapses
- Tetanus toxin has one antigenic type, unlike botulinum toxin, which has eight

Clostridium tetani

62



Clinical Findings

- Tetanus is characterized by strong muscle spasms (**spastic paralysis**, tetany)
- Specific clinical features include **lockjaw** (trismus) due to rigid contraction of the jaw muscles, which prevents the mouth from opening
- **Opisthotonos**, a pronounced arching of the back due to spasm of the strong extensor muscles of the back
- **Respiratory failure**: high mortality rate is associated with this disease
- In tetanus, spastic paralysis (strong muscle contractions) occurs, whereas in botulism, flaccid paralysis (weak or absent muscle contractions) occurs

Clostridium tetani

63

Laboratory Diagnosis

- Clinical diagnosis

Treatment, Prevention, and Control

- Treatment of tetanus requires debridement of the primary wound, use of penicillin or metronidazole to kill the bacteria and reduce toxin production
- passive immunization with human tetanus immunoglobulin to neutralize unbound toxin
- vaccination with tetanus toxoid

Clostridium botulinum

Clostridium botulinum

65

- The etiologic agents of botulism are a heterogeneous collection of large, fastidious, spore-forming, anaerobic rods
- Spores, widespread in soil, contaminate vegetables and meats
- When these foods are canned or vacuum-packed without adequate sterilization, spores survive and germinate in the anaerobic environment
- Toxin is produced within the canned food and ingested preformed
- Disease can be prevented by sufficient cooking

Clostridium botulinum

66

Pathogenesis

- Botulinum toxin
 - absorbed from the gut and carried via the blood to peripheral nerve synapses, where it **blocks release of acetylcholine** → **flaccid paralysis**
 - is a protease that cleaves the proteins involved in acetylcholine release
 - encoded by a lysogenic phage
 - along with tetanus toxin, it is among the most toxic substances known
 - There are eight immunologic types of the toxin; types A, B, and E are the most common in human illness

Clostridium botulinum

67

Clinical Findings

- **Foodborne botulism:** an intoxication resulting from the ingestion of food containing *C. botulinum* toxin
- **Infant botulism:** associated with consumption of foods (e.g., honey, infant milk powder) contaminated with botulinum spores and ingestion of spore-contaminated soil and dust
 - the organisms grow in the gut and produce the toxin there
- **Wound botulism:** spores contaminate a wound, germinate, and produce toxin at the site

Clostridium botulinum

68

Laboratory Diagnosis

- **Clinical**

Treatment, Prevention, and Control

- elimination of the organism from the GI tract using metronidazole or penicillin therapy
- botulinum antitoxin

69

Corynebacterium diphtheria

- Diphtheria

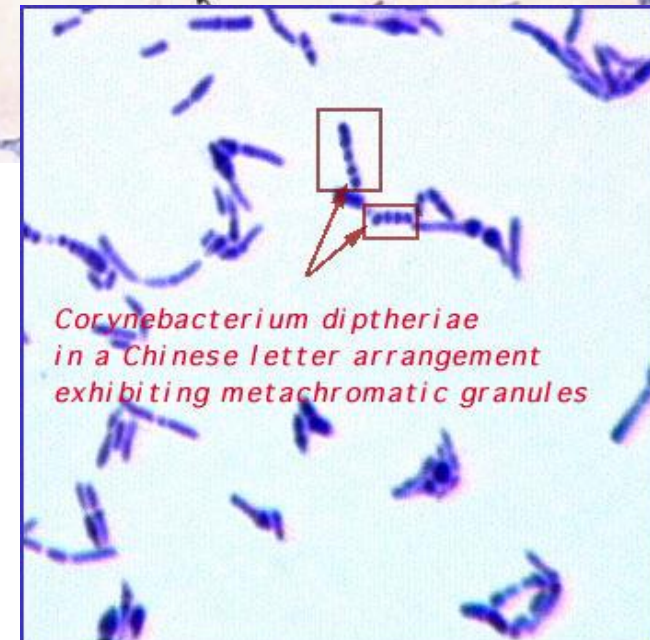
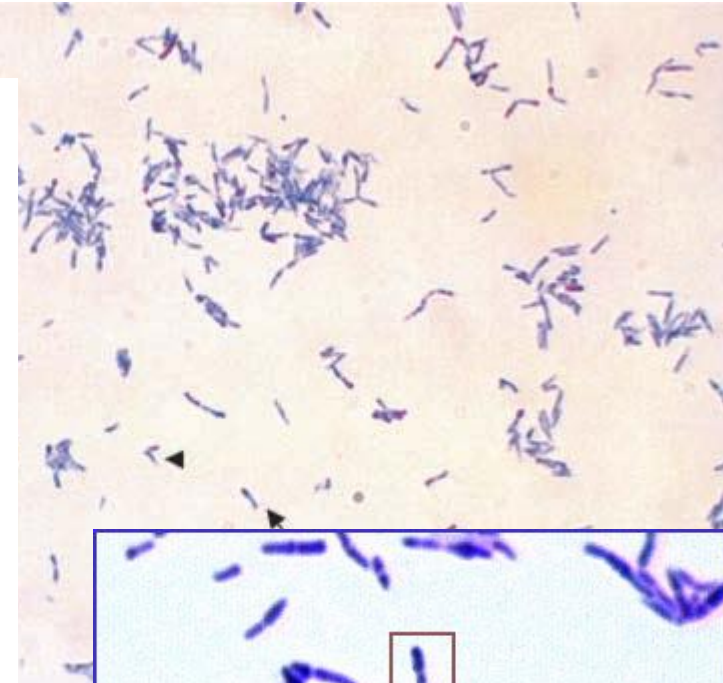
Corynebacterium diphtheriae

70

Properties

- gram-positive rods that appear club-shaped (wider at one end)
- are arranged in V- or L-shaped formations
- are aerobic or facultatively anaerobic, nonmotile, and catalase positive
- The rods have a beaded appearance. The beads consist of granules
- The granules stain metachromatically
- The genus *Corynebacterium* contains 150 species and subspecies

Corynebacterium diphtheriae—Gram stain



Corynebacterium diphtheriae
in a Chinese letter arrangement
exhibiting metachromatic granules

Corynebacterium diphtheriae

71

Transmission and Epidemiology

- Diphtheria is a disease found worldwide
- Humans are the only natural host of *C. diphtheriae*
- *C. diphtheriae* is maintained in the population by asymptomatic carriage in the oropharynx or on the skin of immune people
- **Transmission**
 - by airborne droplets
 - can also infect the skin at the site of a preexisting skin lesion

Corynebacterium diphtheriae

72

Pathogenesis and Immunity

- **Diphtheria toxin**
 - the major virulence factor of *C. diphtheriae*
 - The *tox* gene is introduced into strains of *C. diphtheriae* by a lysogenic bacteriophage, **β-phage**
 - Is an A-B exotoxin
 - **Inhibits protein synthesis** by interfering with elongation factor-2 (EF-2)
 - Affects all eukaryotic cells regardless of tissue type

Corynebacterium diphtheriae

73

Pathogenesis and Immunity

- The host response to *C. diphtheriae* infection
 - 1) A local inflammation in the throat, with a fibrinous exudate that forms the tough, adherent, gray **pseudomembrane** characteristic of the disease
 - 2) Antibody that can neutralize exotoxin activity by blocking the interaction of the binding domain with the receptors, thereby preventing entry into the cell

Corynebacterium diphtheriae

74

Clinical Findings

- The clinical presentation of diphtheria is determined by the
 - Site of infection
 - Immune status of the patient, and
 - Virulence of the organism
- Diphtheria toxin
 - produced at the site of the infection
 - disseminates through the blood to produce the systemic signs of diphtheria

Corynebacterium diphtheriae

75

Clinical Findings

• Respiratory Diphtheria

- Develop after 2- 4 days of incubation period
- Organisms multiply locally on epithelial cells in the pharynx or adjacent surfaces → **exudative pharyngitis**
- The exudate evolves into a thick **pseudomembrane** that can cover the tonsils, uvula, and palate
- The **pseudomembrane firmly adheres to the underlying tissue**



Corynebacterium diphtheriae

76

Clinical Findings

- **Cutaneous Diphtheria**

- acquired through skin contact with infected persons
- The organism colonizes the skin and gains entry into the subcutaneous tissue through breaks in the skin
- A papule develops first and then evolves into a **chronic, nonhealing ulcer**, sometimes covered with a grayish membrane



Corynebacterium diphtheriae

77

Laboratory Diagnosis

- Clinical diagnosis for quick patient management
- Microscopy: Gram stained smear of throat swab
- Culture: Loeffler's medium, a tellurite plate, blood agar plate

Treatment

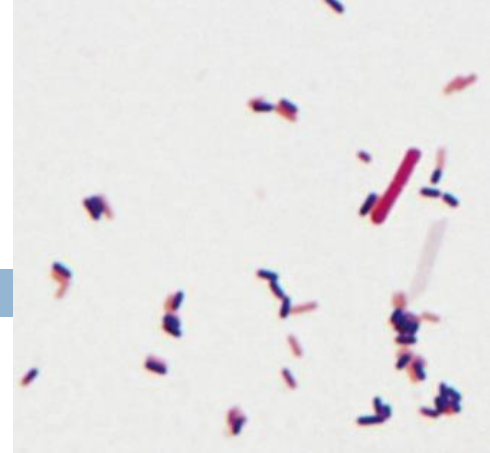
- The treatment of choice is **antitoxin** → to neutralize unbound toxin in the blood
- Treatment with **penicillin G or erythromycin** is also recommended
 - inhibit growth of the organism, reduce toxin production, and decrease the incidence of chronic carriers
- **Prevention:** immunization with diphtheria toxoid (usually given as DTP)

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Listeria monocytogenes

Listeria monocytogenes

79



Properties

- The genus *Listeria* consists of 19 species, with *L. monocytogenes* the most significant human pathogen
- short, non-branching, gram-positive, facultatively anaerobic rod
- capable of growth at a broad temperature range (**1°C to 45°C**) and in a high concentration of salt
- motile at room temperature but less so at 37 °C
- exhibit a characteristic end-over-end **tumbling motion** when a drop of broth is examined microscopically
- exhibits weak **β-hemolysis** when grown on sheep blood agar

Listeria monocytogenes...

80

Pathogenesis and immunity

- *L. monocytogenes* is a facultative intracellular pathogen
- After penetration into the cells, the acid pH of the phagolysosome activates a pore-forming cytolysin (**listeriolysin O**) and two different phospholipase C enzymes
 - leads to **release of the bacteria into the cell cytosol**
- The bacteria proceed to replicate and then move to the cell membrane → adjacent cells ingest the bacterium
- Entry into macrophages after passage through the intestinal lining carries the bacteria to the liver and spleen, leading to disseminated disease

Listeria monocytogenes...

81

Epidemiology

- *L. monocytogenes* is isolated from a variety of environmental sources and from the feces of mammals, birds, fish, and other animals
- The primary source of infection is consumption of contaminated food
- However, human-to-human transmission can occur primarily from mother to child in utero or at birth
- Fecal carriage is estimated to occur in 1-5% of healthy people

Listeria monocytogenes...

82

Epidemiology

- The incidence of disease is common among neonates, the elderly, pregnant women, and patients with severe defects in cell-mediated immunity
- Because *Listeria* can grow in a wide pH range and at cold temperatures, foods with small numbers of organisms can become heavily contaminated during prolonged refrigeration
- Disease can occur if the food is uncooked or inadequately cooked before consumption

Listeria monocytogenes...

83

Clinical Diseases

- **Early-onset neonatal disease**
 - acquired transplacentally in utero
 - can result in abortion, stillbirth, or premature birth
- **Late-onset neonatal disease**
 - acquired at or soon after birth
 - occurs 2 to 3 weeks after birth in the form of meningitis or meningoencephalitis with septicemia
- Pregnant women: nonspecific influenza-like symptoms
- Meningitis in Adults
- Primary Bacteremia

Listeria monocytogenes...

84

Laboratory Diagnosis

• Culture

- *Listeria* grows on most conventional laboratory media, with small, round colonies
- β -Hemolysis on sheep blood agar media
- Characteristic tumbling motility

Treatment and prevention

- combination of gentamicin with either penicillin or ampicillin is the treatment of choice for serious infections
- Vaccine is not available



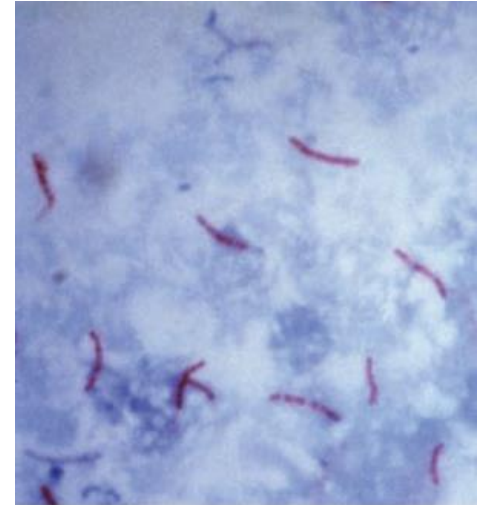
85

Mycobacteria

Mycobacterium tuberculosis

86

- *Mycobacteria* possess a complex, lipid-rich cell wall that is responsible for many of its characteristic properties
 - Acid-fastness; slow growth; resistance to detergents, common antibacterial antibiotics, and the host immune response; antigenicity)
- *M. tuberculosis* grows slowly
 - it has a doubling time of 18 hours
 - Because growth is so slow, cultures of clinical specimens must be held for 6 to 8 weeks before being recorded as negative



M. tuberculosis...

87

- Weakly gram-positive, strongly acid-fast rods
- *M. tuberculosis* is an **obligate aerobe**
 - its predilection for causing disease in highly oxygenated tissues such as the upper lobe of the lung and the kidney
- The acid-fast property of *M. tuberculosis* is attributed to long-chain (**C78–C90**) fatty acids called **mycolic acids** in the cell wall
- Cord factor (trehalose dimycolate) is correlated with virulence of the organism

M. tuberculosis...

88

Transmission & Epidemiology

- *M. tuberculosis* is transmitted from person to person by **respiratory aerosols** produced by coughing
- The source of the organism is a cavity in the lung that has eroded into a bronchus
- The portal of entry is the respiratory tract, and the initial site of infection is the lung
- In tissue, it resides chiefly within reticuloendothelial cells (e.g., macrophages)

M. tuberculosis...

Transmission & Epidemiology

- Humans are the natural reservoir of *M. tuberculosis*
- Although some animals, such as cattle, can be infected, they are not the main reservoir for human infection
- In developing countries, *M. bovis* also causes tuberculosis in humans
- *M. bovis* is found in cow's milk, which, unless pasteurized, can cause gastrointestinal tuberculosis in humans

M. tuberculosis...

90

Transmission & Epidemiology

- Tuberculosis occurs in only a small number of infected individuals
- Populations at greatest risk for disease are
 - immunocompromised patients (particularly those with HIV infection)
 - drug or alcohol abusers
 - homeless persons
 - individuals exposed to diseased patients

M. tuberculosis...

91

Transmission & Epidemiology

- The WHO estimated that **one-fourth of the world's population** is infected with *M. tuberculosis*
- In 2019
 - an estimated 10 million people fell ill with tuberculosis worldwide
 - a total of 1.4 million people died from TB

M. tuberculosis...

Pathogenesis and Immunity

- *M. tuberculosis* is an intracellular pathogen able to establish lifelong infection
- At the time of exposure, *M. tuberculosis* enters the respiratory airways
- Few of them penetrate to the alveoli, where they are phagocytized by alveolar macrophages

M. tuberculosis...

93

Pathogenesis and Immunity

- *M. tuberculosis* survives and multiplies within a cellular vacuole called a phagosome
- The organism produces a protein called exported repetitive protein
 - prevents the phagosome from fusing with the lysosome
 - allow the organism to escape the degradative enzymes in the lysosome

M. tuberculosis...

Clinical Diseases

- Although tuberculosis can involve any organ, most infections in immunocompetent patients are restricted to the lungs
- The patient's cellular immunity is activated and mycobacterial replication ceases in most patients within 3 to 6 weeks after exposure to the organism
- Approximately 5% of patients exposed to *M. tuberculosis* progress to having active disease within 2 years
- 5% to 10% experience disease sometime later in life
- The likelihood that infection will progress to active disease is a function of both the infectious dose and the patient's immune competence

M. tuberculosis...

95

Clinical Diseases

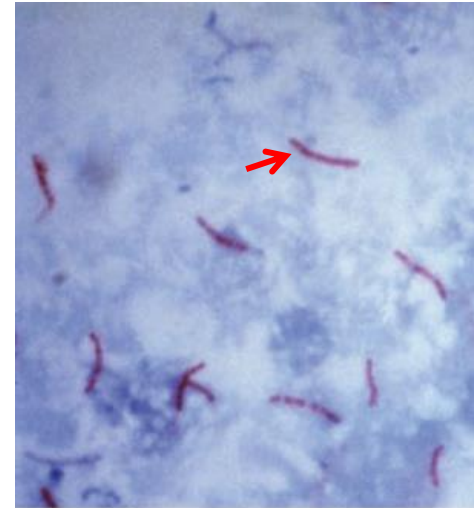
- **Pulmonary tuberculosis**
 - Patients typically have nonspecific complaints of malaise, weight loss, cough, and night sweats
 - Sputum may be scant or bloody and Purulent
- **Extrapulmonary tuberculosis**
 - occur as the result of the hematogenous spread of the bacilli during the initial phase of multiplication

M. tuberculosis...

96

Diagnosis

- Tuberculin skin test and interferon (IFN)- γ release tests
 - sensitive markers for exposure to the organism
- Microscopy: Acid fast staining
- Culture: gold standard, but take longer
- Nucleic acid amplification tests
 - PCR
 - Xpert MTB/RIF
- Identification most commonly made using species-specific molecular probes, sequencing, or mass spectrometry



M. tuberculosis...

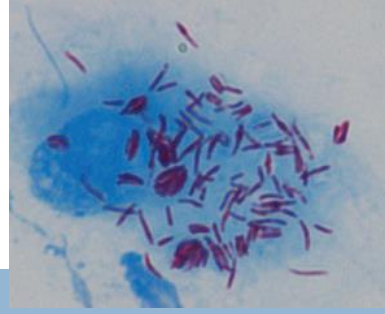
Treatment, Prevention, and Control

- Prolonged treatment with multiple drugs is required to prevent development of drug-resistant strains
- **Isoniazid (INH), ethambutol, pyrazinamide, and rifampin for 2 months followed by 4-6 months of INH and rifampin or alternative combination drugs**
- Control of disease through
 - active surveillance
 - prophylactic and therapeutic intervention
 - careful case monitoring

Mycobacterium leprae

Leprosy (*Hansen's disease*)

Mycobacterium leprae



99

Properties

- First identified by the Norwegian physician Gerhard Armauer Hansen in 1873
- acid-fast, non-motile, non-spore, weakly gram-positive, obligate intracellular bacillus that cannot be cultivated in vitro
- Less acid fast than MTB → 1% acid alcohol is used as a decolorizer
- Consists of a Lipid-rich cell wall (60% lipid) → responsible for acid fastness, resistance to disinfectants, detergents, common antibacterial antibiotics, and traditional stains
- Grows best at temperatures b/n 27°C and 30°C
- infect preferably the cold areas of the body such as the skin, nasal mucosa and peripheral nerves
- Grows very slowly: approximate generation time of 12–14 days

Mycobacterium leprae

100

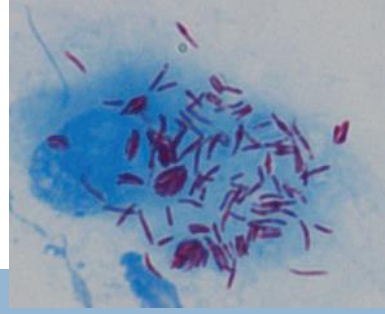


armadillo

Leprosy

- chronic infectious disease caused by *Mycobacterium leprae*
- mainly affects the skin and nerves
- progresses slowly with an average incubation period of 3 years
- Leprosy can affect all ages and both sexes
- leprosy is also endemic in armadillos
- Leprosy can be cured → Multi-Drug therapy (MDT) kills the bacteria and stops the spread of the disease
- If detected early and treated with MDT, leprosy will not lead to disabilities

Mycobacterium leprae

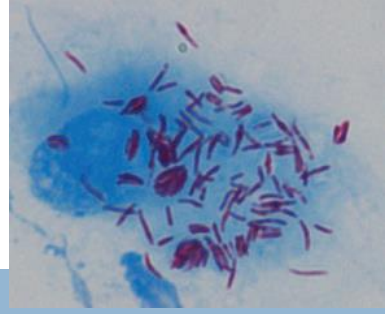


101

Epidemiology and transmission

- Leprosy transmission is not entirely understood
- *M. leprae* is not very virulent → most people affected with leprosy are non-infectious → bacilli remain within the infected cells
- Multibacillary patients, however, excrete *M. leprae* from their nasal mucosa and skin
- long-term exposure of the respiratory system to airborne droplets is the main route of infection
- Close and repeated contact with these patients is also a source of transmission

Mycobacterium leprae



102

Epidemiology and transmission

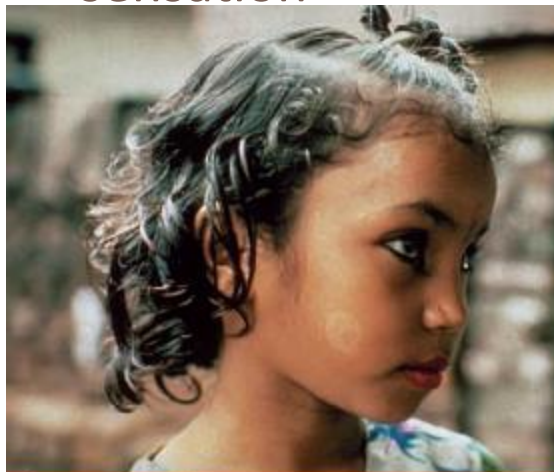
- *M. leprae* primarily infects
 - histiocytes (or tissue macrophages) in the dermis
 - Schwann cells in the peripheral nerves
- Once it penetrates a cell, replication proceeds slowly until T cells recognize the mycobacterial antigens and a chronic inflammatory reaction begins
- The outcome of infection and clinical manifestation depend on the cellular immunity of the host
- *M. leprae* escape macrophage killing by Inhibiting phagosome-lysosome fusion

Mycobacterium leprae

103

Clinical features

- Leprosy is characterized by skin patch or patches with a definite loss of sensation



Leprosy patches...

...can be pale or reddish or copper-coloured.

...can be flat or raised.



Leprosy patches...

...usually do not hurt.



...lack sensation to heat, touch or pain.

...can appear anywhere.



...do not itch.



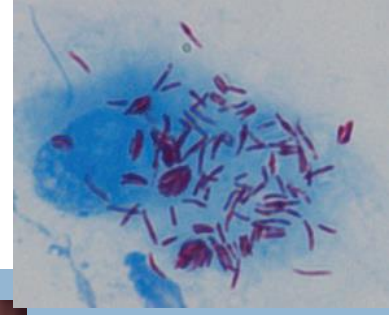
Leprosy can be diagnosed on clinical signs alone.



Leprosy...Clinical features...

104

Facial nerve paralysis



- Leprosy with peripheral nerve damage



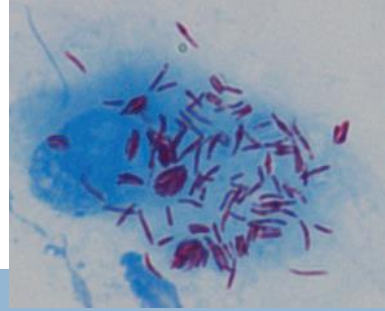
Dropped wrist, clawed fingers with stiff joints due to ulnar and median nerve damage



Foot ulceration due to loss of sensation

Classification of leprosy

105



a. **Tuberculoid leprosy**

- milder and is characterized by hypopigmented skin macules
- have a strong cellular immune reaction, relatively few bacteria
- cytokine production → macrophage activation → phagocytosis → bacillary clearance

b. **Lepromatous leprosy**

- associated with disfiguring skin lesions, nodules, thickened dermis, and involvement of the nasal mucosa
- strong antibody response, defected cellular response, abundance of bacteria
- most infectious form of leprosy

Based on number of patches

a. Paucibacillary (PB) leprosy

- 1–5 patches

b. Multibacillary (MB) leprosy

- More than 5 patches

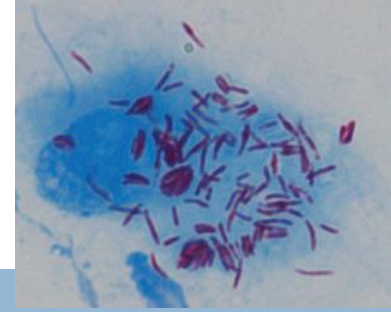


MB



Leprosy...Diagnosis

106



Clinical diagnosis

- skin patch or patches with a definite loss of sensation

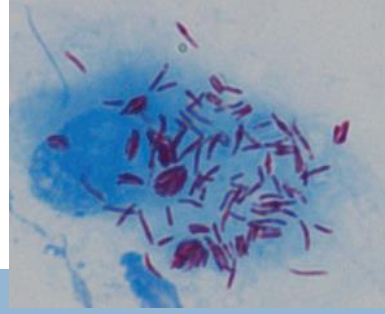
How to Test for Sensory Loss

Take a pointed object such as a pen.
Show the person what you are going to do.
Lightly touch the skin with the pen.
Ask the person to point to where they felt the pen.
Now ask them to close their eyes so that they cannot see what you are doing.
Lightly touch the centre of the most prominent skin patch and ask them to point to where they felt the pen.
Repeat the procedure on normal skin and on the same patch again.
If the person feels nothing on the skin patch, it is leprosy. Start treatment immediately.



Generally leprosy should not be diagnosed without a definite loss of sensation.

Leprosy...Laboratory diagnosis



107

Skin smear

- Has specificity of 100% and a sensitivity of 50%
- Occasionally needed to confirm a clinical diagnosis of multibacillary leprosy
- Specimen: taken from the edge of a leprosy lesion, nasal mucosa, an ear lobe
- For detail ([Monica Part 2, 2nd ed: pages 212-216](#))

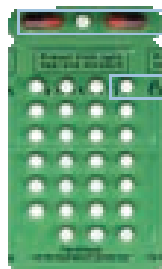
PCR: highly specific and sensitive method, although expensive

Leprosy...treatment

108

MDT Regimens

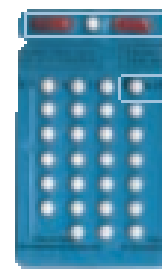
Each blister pack contains treatment for 4 weeks.



PB adult blister pack

PB adult treatment:

- Once a month: Day 1
 - 2 capsules of rifampicin (300 mg X 2)
 - 1 tablet of dapsone (100 mg)
- Once a day: Days 2-28
 - 1 tablet of dapsone (100 mg)
- Full course: 6 blister packs

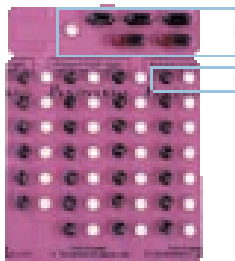


PB child blister pack

PB child treatment (10-14 years):

- Once a month: Day 1
 - 2 capsules of rifampicin (300 mg+150 mg)
 - 1 tablet of dapsone (50 mg)
- Once a day: Days 2-28
 - 1 tablet of dapsone (50 mg)
- Full course: 6 blister packs

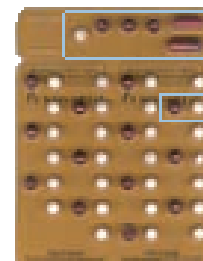
For children younger than 10, the dose must be adjusted according to body weight.



MB adult blister pack

MB adult treatment:

- Once a month: Day 1
 - 2 capsules of rifampicin (300 mg X 2)
 - 3 capsules of clofazimine (100mg X 3)
 - 1 tablet of dapsone (100 mg)
- Once a day: Days 2-28
 - 1 capsule of clofazimine (50 mg)
 - 1 tablet of dapsone (100 mg)
- Full course: 12 blister packs

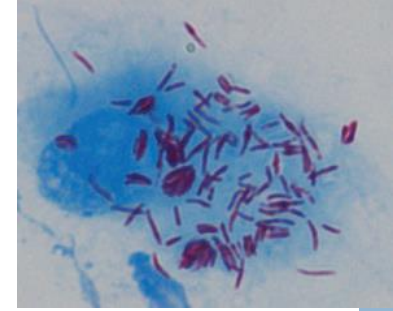


MB child blister pack

MB child treatment (10-14 years):

- Once a month: Day 1
 - 2 capsules of rifampicin (300 mg+150 mg)
 - 3 capsules of clofazimine (50 mg X 3)
 - 1 tablet of dapsone (50 mg)
- Once a day: Days 2-28
 - 1 capsule of clofazimine every other day (50 mg)
 - 1 tablet of dapsone (50 mg)
- Full course: 12 blister packs

For children younger than 10, the dose must be adjusted according to body weight.



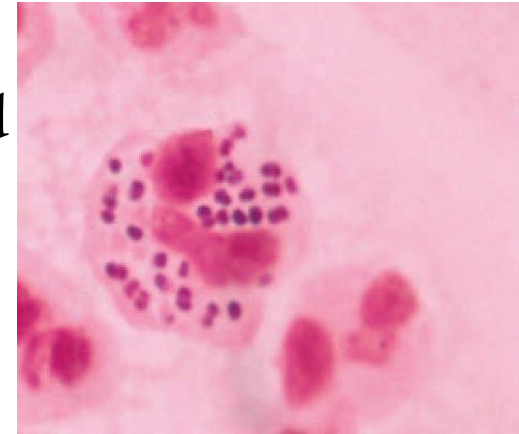
It is crucial
that patients understand
which drugs they have
to take once a month and which
every day.

Gram Negative Bacteria

Neisseria

Physiology and Structure

- Gram-negative diplococci resembling paired kidney beans
- Neisseriae are oxidase-positive
- Grow best at 35°C to 37°C in a humid atmosphere supplemented with CO₂
 - Chocolate agar
- Two medically important species that can infect human are *N. gonorrhoeae*, *N. meningitides*
- lipooligosaccharide (LOS) induce inflammatory reaction → causes most gonococcal diseases symptoms



Neisseria gonorrhoeae (gonococcus)

111

- has no polysaccharide capsule unlike *N. meningitidis*
- has multiple serotypes based on the antigenicity of its pilus protein
- There is marked antigenic variation in the gonococcal pili → more than 100 serotypes are known
- Gonococci have three outer membrane proteins (proteins I, II, and III)
- Protein II plays a role in attachment of the organism to cells and varies antigenically

Neisseria gonorrhoeae (gonococcus)

112

- Cause disease in humans only

Transmission

- primarily by sexual contact
 - Because gonococcus is quite sensitive to dehydration and cool conditions, sexual transmission favors its survival
- newborns can be infected during birth
- Genital tract infections are the most common source of the organism, but **anorectal and pharyngeal infections are important sources as well**

Neisseria gonorrhoeae (gonococcus)

113

Virulence factors

A. Pili

- most important virulence factors → mediate attachment to mucosal cell surfaces and are antiphagocytic
- Piliated gonococci are usually virulent, whereas nonpiliated strains are avirulent

B. Endotoxin (lipooligosaccharide, LOS)

C. IgA protease

- hydrolyze secretory IgA, which could otherwise block attachment to the mucosa

Neisseria gonorrhoeae (gonococcus)

114

Pathogenesis and clinical features

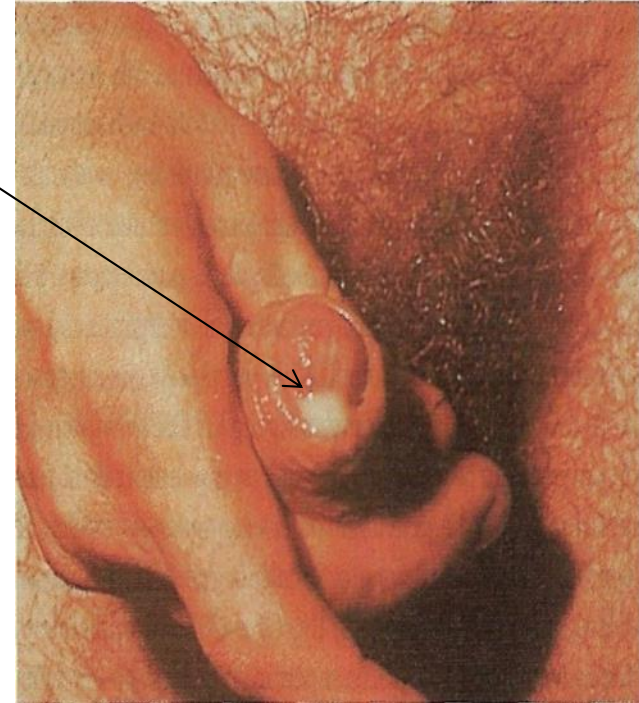
- Gonococci infect primarily the mucosal surfaces (e.g., the urethra and vagina), but dissemination occurs
- Gonococci cause both localized and disseminated infections with seeding of various organs
 - Gonococci reach these organs via the bloodstream (gonococcal bacteremia)
- Gonorrhea is usually symptomatic in men but often asymptomatic in women
 - In men primarily restricted to the urethra
 - The primary site of infection in women is the cervix

Neisseria gonorrhoeae (gonococcus)

115

Pathogenesis and clinical features

- Gonorrhea in men
 - characterized primarily by urethritis accompanied by dysuria and a purulent discharge after 2 to 5 days incubation period
 - Epididymitis can occur



Neisseria gonorrhoeae (gonococcus)

116

Pathogenesis and clinical features

In women

- infection is located primarily in the endocervix
- cause a **purulent vaginal discharge and intermenstrual bleeding (cervicitis)**
- The most frequent **complication** in women is an **ascending infection of the uterine tubes** (salpingitis, PID)
 - can result in **sterility or ectopic pregnancy** as a result of scarring of the tubes

Neisseria gonorrhoeae (gonococcus)

117

Pathogenesis and clinical features

In newborn infants

- Purulent conjunctivitis (ophthalmia neonatorum)
- Result of gonococcal infection acquired from the mother during passage through the birth canal
- Gonococcal conjunctivitis also occurs in adults as a result of the transfer of organisms from the genitals to the eye

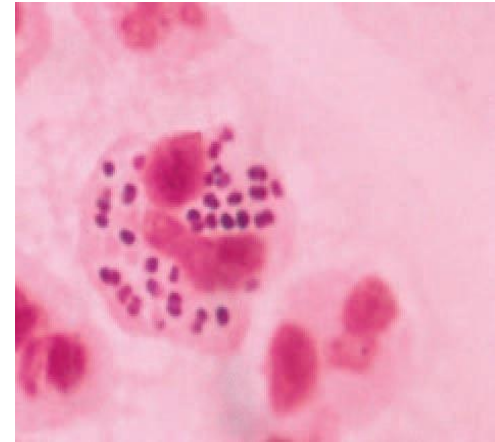


Neisseria gonorrhoeae (gonococcus)

118

Laboratory Diagnosis

- **Specimen:** Urethral discharge from male, Cervical swab from female
- **Microscopy:** Gram negative intracellular diplococci
- **Culture:** Thayer-Martin medium, which is a chocolate agar containing antibiotics (vancomycin, colistin, trimethoprim, and nystatin) to suppress the normal flora



Neisseria gonorrhoeae (gonococcus)

119

Treatment

- Ceftriaxone is currently the treatment of choice

Prevention

- use of condoms
- the prompt treatment of symptomatic patients and their sex partners
- Gonococcal conjunctivitis in newborns: erythromycin ointment

Neisseria meningitidis

120

- Humans are the only natural hosts for meningococcus
- Transmitted by **airborne droplets**
- Colonize the membranes of the nasopharynx and become part of the transient flora of the upper respiratory tract
- From the nasopharynx, the organism can enter the bloodstream and spread to specific sites, such as the meninges or joints, or be disseminated throughout the body (meningococemia)
- About 5% of people become chronic carriers and serve as a source of infection for others

Neisseria meningitidis...

121

- Overall, *N. meningitidis* ranks second to *S. pneumoniae* as a cause of meningitis
- But it is the most common cause in persons between the ages of 2 and 18 years

Neisseria meningitidis...

122

Virulence factors

1. Polysaccharide capsule

- Enables the organism to resist phagocytosis
- The capsule is the immunogen
- Basis for serogrouping: 6 serogroups (A, B, C, W135, X, and Y)

2. Endotoxin:

- Causes fever, shock, and other pathophysiologic changes

3. IgA protease

- helps the bacteria attach to the membranes of the upper respiratory tract by **cleaving secretory IgA**

4. Factor H binding protein (FHBP) on meningococci

- binds Factor H→inhibitor of complement factor C3b

Neisseria meningitidis...

123

Clinical Findings

- The two most important manifestations of disease are **meningococemia** and **meningitis**
- Bacteremia can result in the seeding of many organs, especially the meninges
- The symptoms of meningococcal meningitis are those of typical bacterial meningitis
 - Fever, headache, stiff neck, and an increased level of PMNs in the spinal fluid

Neisseria meningitidis...

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Laboratory Diagnosis

- Specimen: Blood, CSF
- Microscopy: Gram negative intracellular diplococci
- Culture: Thayer-Martin medium, which is a chocolate agar containing antibiotics (vancomycin, colistin, trimethoprim, and nystatin) to suppress the normal flora

Treatment

- Either ceftriaxone or penicillin G is the drug of choice

Prevention: Meningococcal Vaccines

Serogroups Covered	Immunogen	Carrier Protein	Where Available	Name of Vaccine
A, C, Y, W-135	Capsular polysaccharide	Diphtheria toxoid	United States	Menacta, Menveo
A, C, Y, W-135	Capsular polysaccharide	None	Countries other than United States	Menomune
A	Capsular polysaccharide	Diphtheria toxoid	Africa's meningitis belt	MenAfriVac
C, Y plus <i>Haemophilus influenzae</i>	Capsular polysaccharide	Tetanus toxoid	United States	MenHibrix
B	Factor H binding protein	None	United States	Trumemba
B	Factor H binding protein, NadA, NHBA, OMV	None	United States	Bexsero

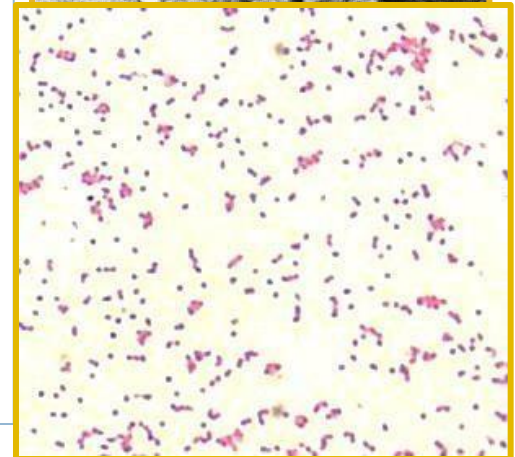
Brucellae



- Gram-negative, non-encapsulated coccobacilli (0.6-0.8 μ m)
- first isolated by Bruce in 1887
- comprises 10 species according to their host specificities
- 4 species are the main pathogens of human and livestock
 - *B. melitensis*, *B. abortus*, *B. suis*, *B. canis*
- Transmission
 - ingestion of contaminated and unpasteurized milk
 - Occupational contact with infected animals
 - Inhalation



Sir David Bruce
(1855-1931)



Brucellosis...clinical features

Human

- Asymptomatic infection
- Symptomatic infection
 - Acute disease develops in approximately half of the patients
 - Incubation period → 1 to 3 weeks
 - **Initial symptoms** (nonspecific): malaise, chills, sweats, fatigue, weakness
 - Almost all patients have fever, and this can be intermittent in untreated patients → **undulant fever**
 - Chronic infections can develop in inadequately treated patients

Animals

- sterility, abortions, or asymptomatic lifelong carriage

Brucellosis...diagnosis

- Culture
 - Blood, bone marrow, other tissues
 - Blood cultures should be incubated for 2 weeks before they are considered negative
- Serology (antibody detection)
- Immunofluorescence
 - Organism in clinical specimens
- PCR

Bordetella pertussis

- Small encapsulated gram-negative rods
- Disease: Whooping cough (pertussis)
- Transmission: respiratory droplets
- Virulence factors
 - **Pertussis toxin:** stimulates adenylate cyclase → impairs leukocyte chemotaxis → inhibits phagocytosis and causes local edema
 - **Tracheal cytotoxin:** damages ciliated epithelium of respiratory tract

Bordetella pertussis...

Table II-2-19. Stages of Whooping Cough (Pertussis) vs. Results of Bacterial Culture

	Incubation	Catarrhal	Paroxysmal	Convalescent
Duration	7-10 days	1-2 weeks	2-4 weeks	3-4 weeks (or longer)
Symptoms	None	Rhinorrhea, malaise, sneezing, anorexia	Repetitive cough with whoops, vomiting, leukocytosis	Diminished paroxysmal cough, development of secondary complications (pneumonia, seizures, encephalopathy)
Bacterial Culture				

Bordetella pertussis...

Lab Diagnosis

- Fastidious/delicate: Regan-Lowe or Bordet-Gengou media; either direct cough plates or nasopharyngeal cultures
- Direct immunofluorescence (DFA) on nasopharyngeal smear
- PCR
- serologic tests

Treatment

- Supportive care
- erythromycin for 14 days including all household contacts

Prevention: Vaccine

Haemophilus influenzae

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- small, pleomorphic gram-negative rods with a polysaccharide **capsule**
- one of the three important encapsulated pathogens, along with the pneumococcus and the meningococcus
- Serologic typing is based on the antigenicity of the capsular polysaccharide
- Six serotypes (a–f), type b is the most important
 - Type b used to cause most of the severe, invasive diseases, such as meningitis and sepsis

Haemophilus influenzae

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- Their growth requires supplementation of media with two growth-stimulating factors
 - Hemin (X factor for “unknown factor”)
 - Nicotinamide adenine dinucleotide (NAD) [V factor for “vitamin”]
 - Chocolate agar is used for the isolation of *Haemophilus* in culture

Haemophilus influenzae

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Pathogenesis & Epidemiology

- *H. influenzae* infects only humans; no animal reservoir
- Transmission: **inhalation of airborne droplets** into the respiratory tract
- Most infections occur in children between the ages of 6 months and 6 years, with a peak in the age group from 6 months to 1 year
- **Virulence factors**
 - Capsule
 - Endotoxin
 - IgA protease: degrades secretory IgA, thus facilitating attachment to the respiratory mucosa

Haemophilus influenzae

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Clinical Diseases

- Meningitis
 - *H. influenzae* type *b* was the most common cause of pediatric meningitis
 - results from bacteremic spread of the organisms from the nasopharynx
- Sinusitis
- Otitis media
- septic arthritis, cellulitis, and sepsis

Haemophilus influenzae

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Laboratory Diagnosis

- Sample: Blood, cerebrospinal fluid
- Isolation on chocolate agar enriched with two growth factors required for bacterial respiration
 - factor X and factor V
- An organism that grows only in the presence of both growth factors is presumptively identified as *H. influenzae*



Haemophilus influenzae

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Treatment

- Meningitis: ceftriaxone
- otitis media and sinusitis: amoxicillin-clavulanate ,
trimethoprim-sulfamethoxazole

Prevention

- Vaccine contains the capsular polysaccharide of *H. influenzae* type b conjugated to diphtheria toxoid

Haemophilus ducreyi

- *H. ducreyi* causes **soft chancre** or **chancroid**
- Chancroid
 - Sexually transmitted disease
 - most commonly diagnosed in men
 - asymptomatic or inapparent in women
 - Approximately 5 to 7 days after exposure, a **tender papule** with an erythematous base develops on the genitalia or perianal area
 - Within 2 days the **lesion ulcerates** and becomes painful, and **inguinal lymphadenopathy** is commonly present
 - Other causes of genital ulcers: syphilis and herpes simplex



Source: Goldsmith LA, Katz SL, Gilchrist RA, Paller AS, Leffell DJ, Wolff K: Fitzpatrick's Dermatology in General Medicine, 8th edition: www.accessmedicine.com

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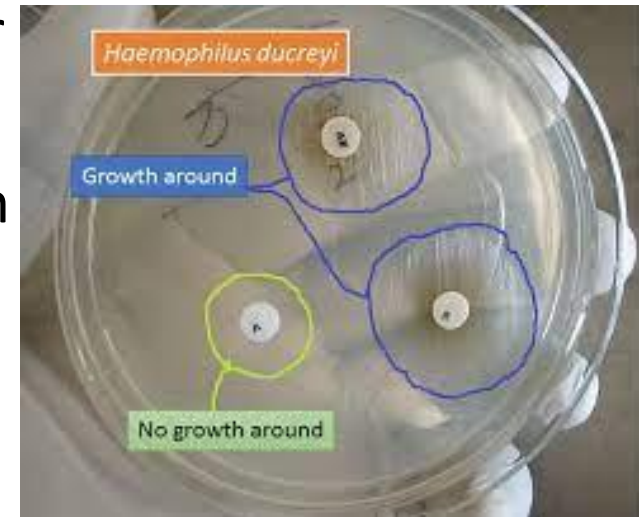
Haemophilus ducreyi...

Laboratory Diagnosis

- isolating *H. ducreyi* from the ulcer (swab) or from pus aspirated from a lymph node
- Requires chocolate agar supplemented with X factor (heme) but, unlike *H. influenzae*, does not require V factor (NAD)

Treatment

- erythromycin, azithromycin, or ceftriaxone
- Because many strains of *H. ducreyi* produce a plasmid-encoded penicillinase, penicillins cannot be used



Vibrio

- Currently the genus *Vibrio* is composed of 76 species of motile curved rods
- *Vibrio cholerae*, *Vibrio parahaemolyticus*, and *Vibrio vulnificus* are medically important
- There are 140 serogroups of *V. cholerae* (O1-O140)
- *V. cholerae* O1 and O139 produce cholera toxin and are associated with epidemics of cholera
- Spread by consumption of contaminated food or water
- Direct person-to-person spread is rare because the infectious dose is high

Vibrio...

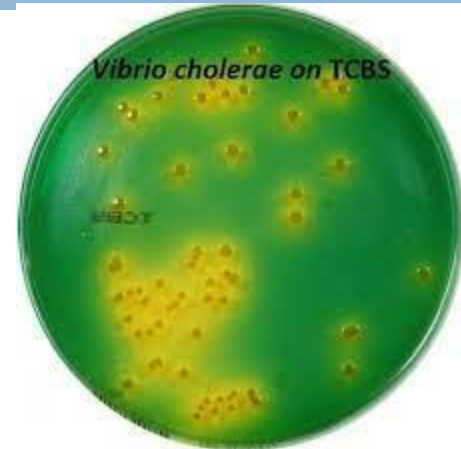
Clinical Diseases

- **majority** of individuals exposed to toxigenic *V. cholerae* O1 have **asymptomatic** infections or self-limited diarrhea
- some individuals develop severe, rapidly fatal diarrhea
- clinical manifestations of cholera begin an average of 2 to 3 days after ingestion
- abrupt onset of watery diarrhea ("rice-water" stools) and vomiting
- The mortality rate is 60% in untreated patients but less than 1% in treated

Vibrio...

Laboratory diagnosis

⇒ Stool culture: TCBS agar



Treatment

⇒ Fluid and electrolyte replacement are crucial

⇒ Antibiotics (e.g., azithromycin) reduce the bacterial burden and exotoxin production, as well as duration of diarrhea

Campylobacter

- Small (0.2-0.5 μm in diameter), S-shaped, motile (single polar flagellum) gram-negative rods
- Microaerophilic: grow best in an atmosphere of reduced oxygen (5% to 7%), increased carbon dioxide (5% to 10%) and at 42°C
- Lipopolysaccharide is the major antigen



Campylobacter...

- There are 25 species; 2 are significant human pathogens
 - *Campylobacter jejuni*
 - *Campylobacter coli*
- *Campylobacter* infections are zoonotic, with a variety of animals serving as reservoirs; i.e. cattle, sheep, rodents, poultry, pig, dogs, cats etc.
- Infections are acquired by consumption of contaminated food, especially poultry, milk or water
- Person to person spread by the fecal-oral route is rare

Campylobacter...

Virulence factors

- LPS: induce inflammatory response
- Enterotoxin: watery diarrhea

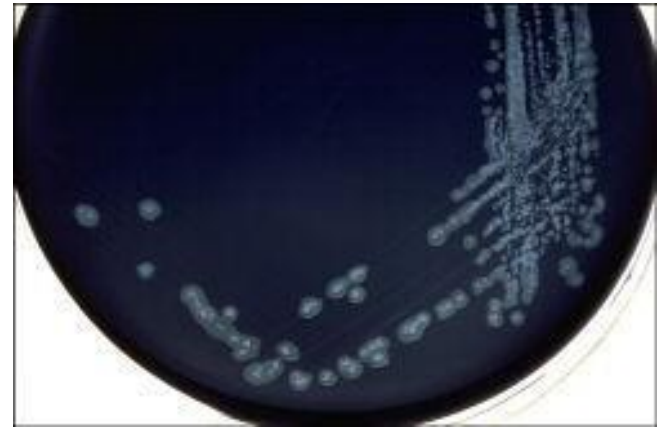
Clinical features

- Asymptomatic infection
- Diarrhea

Laboratory diagnosis

- ⇒ Stool microscopy (Gram staining): "S-shaped" gram negative bacteria
- ⇒ **Stool culture:** *C. jejuni*, *C. coli*, and *C. upsaliensis* require culture at 10% CO₂ and 42°C

Treatment: Erythromycin or azithromycin



Helicobacter pylori

- ⇒ Gram-negative, spiral rods resembling *campylobacter*
- ⇒ highly motile (corkscrew motility) via polar flagella
- ⇒ Produce an abundance of urease
- ⇒ catalase and oxidase-positive
- ⇒ Require microaerophilic conditions (decreased oxygen and increased carbon dioxide); and a temperature range between 30° C and 37° C
- ⇒ Infections are common, particularly in people in a low socioeconomic class (70-90% colonized)
- ⇒ Humans are the primary reservoir (primates, pigs)
- ⇒ colonization is believed to persist for life unless the host is specifically treated
- ⇒ Person-to-person spread is important (typically fecal-oral)

Helicobacter pylori

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Virulence factors

- Initial colonization is facilitated by
 - blockage of acid production by a bacterial acid-inhibitory protein
 - neutralization of gastric acids by the ammonia produced by bacterial urease activity
- Vacuolating cytotoxin A (VacA)
 - damages epithelial cells by producing vacuoles
- cytotoxic-associated gene (CagA)
 - injected in to epithelial cells through type IV secretion system
→ interferes with the normal cytoskeletal structure of the epithelial cells
- cag PAI (phosphoribosylanthranilate isomerase)
 - induce interleukin-8 (IL-8) production → attracts neutrophils

Helicobacter...clinical features

Acute gastritis:

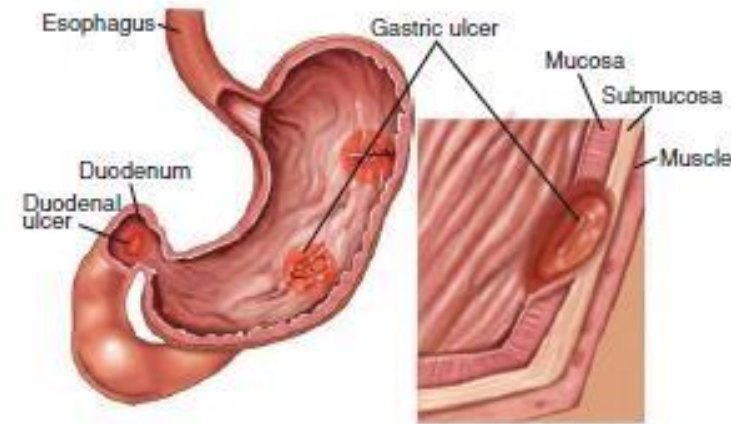
- characterized by a feeling of fullness, nausea, vomiting
- This can evolve into chronic gastritis

Chronic gastritis:

- may involve the entire stomach (pangastritis)
- 10% to 15% of patients with chronic gastritis will progress to develop **peptic ulcers**
- peptic ulcers → gastric ulcer, duodenal ulcer

Cancer:

- cagA-positive strains are associated with a higher risk for gastric cancer



Helicobacter...Laboratory Diagnosis...

⇒ Antigen Detection

- to detect *H. pylori* in stool

⇒ Urea breath test

- Urease testing → following consumption of an isotopically labeled urea solution

⇒ Culture

- *H. pylori* adheres to gastric mucosa =not recovered in stool
- enriched medium supplemented with blood, hemin, or charcoal and incubated in a **microaerophilic atmosphere for up to 2 weeks** = oxidase, catalase, and **urease** positive

Serology: can't discriminate between past and current infection

Helicobacter...

Treatment

- A greatest success in curing gastritis or peptic ulcer disease has been accomplished with the combination of
 - a proton pump inhibitor (e.g., [omeprazole](#)), a macrolide (e.g., [clarithromycin](#)), and a beta-lactam (e.g., [amoxicillin](#)) for 2 weeks
 - Treatment failure is most commonly associated with clarithromycin resistance
 - Metronidazole can also be used in combination therapy, but resistance is common

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Enterobacteriaceae

Enterobacteriaceae

Physiology and Structure

- ☐ Gram-negative non-spore forming rods
- ☐ share a common antigen (enterobacterial common antigen)
- ☐ facultative anaerobes
- ☐ ferment glucose, reduce nitrate, and are catalase positive and **oxidase negative**

Medically Important Enterobacteriaceae

- ❑ *Citrobacter freundii*, *Citrobacter koseri*
- ❑ *Escherichia coli*
- ❑ *Klebsiella pneumoniae*, *Klebsiella oxytoca*
- ❑ *Morganella morganii*
- ❑ *Proteus mirabilis*
- ❑ *Salmonella enterica*
- ❑ *Serratia marcescens*
- ❑ *Shigella sonnei*, *Shigella flexneri*
- ❑ *Yersinia pestis*, *Yersinia enterocolitica*, *Yersinia pseudotuberculosis*

Salmonella

Characteristics and classification

- Motile, never ferment lactose or sucrose, usually produce H_2S
- most clinically significant isolates belong to the species *Salmonella enterica*
- There are more than 2500 unique serotypes *Salmonella enterica*
- **Medically important serotypes:** *Salmonella* Typhi, *Salmonella* Paratyphi, *Salmonella* Choleraesuis

Salmonella...

Epidemiology

- *Salmonella* can colonize virtually all animals
- *S. Typhi* and *S. Paratyphi* are strict human pathogens
- *Salmonella Choleraesuis* is adapted to animals and can cause severe disease when infect humans
- *S. Typhi* and *S. Paratyphi* can survive in the gall bladder and establish chronic carriage
- Infection is acquired through ingestion of contaminated poultry, eggs, dairy products, and foods

Salmonella...

Clinical Diseases

A. Asymptomatic Colonization

B. Gastroenteritis

- most common form of salmonellosis
- Symptoms generally appear 6 to 48 hours after ingestion
- nausea, vomiting, nonbloody diarrhea, Fever, abdominal cramps, myalgias, and headache

C. Enteric Fever (typhoid fever)

- Caused by *S. Typhi* (sever) and *S. Paratyphi* (mild)

Salmonella...

Laboratory diagnosis

- ☐ Stool culture (gastroenteritis, typhoid fever)
- ☐ Blood culture (typhoid fever)
- ☐ Serologic test (widal test): less important

Treatment

- ☐ ciprofloxacin, trimethoprim-sulfamethoxazole, or a broad-spectrum cephalosporin

Shigella

- Nonmotile, non lactose fermenter, fail to produce gas when fermenting glucose
- *S. dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei* are medically important
- Humans are only reservoir for these bacteria
- Disease spread person to person by fecal-oral route
- Relatively few organisms can produce disease (highly infectious)
- *S. dysenteriae* produce **Shiga toxin** w/c inhibit protein synthesis leading to intestinal epithelium damage

Shigella...

Clinical Diseases

- ☐ Gastroenteritis (shigellosis): an initial watery diarrhea progressing within 1 to 2 days to abdominal cramps and tenesmus
- ☐ Bacillary dysentery: caused by *S. dysenteriae*

Laboratory diagnosis

- ☐ Stool culture

Treatment

- ☐ Fluoroquinolone, trimethoprim-sulfamethoxazole

Spirochetes

- The spirochetes are a large, heterogeneous group of spiral, motile gram-negative bacteria
- The family Treponemataceae includes three genera whose members are human pathogens
 - a. Treponema*
 - b. Borrelia*
 - c. Leptospira*

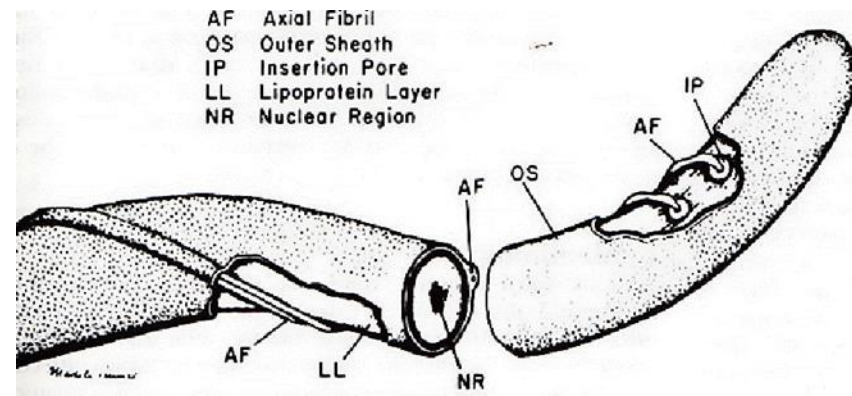
Treponema pallidum

- Syphilis (ཕྱུ་རྒྱུ་)

Treponema pallidum

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- thin, tightly coiled spirochetes with pointed, straight ends
- too thin to be seen with Gram or Giemsa stains
- motile through the undulation of axial filaments (3) that lie under the outer sheath
- *Treponema pallidum* has not been grown on bacteriologic media or in cell culture



Treponema pallidum

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Transmission & Epidemiology

- *Treponema pallidum* is transmitted
 - from spirochete-containing lesions of skin or mucous membranes (e.g., genitalia, mouth, and anus) of an infected person to other persons by intimate contact
 - from pregnant women to their fetuses
 - blood for transfusions collected during early syphilis
- *T. pallidum* is a human organism only
- Syphilis occurs worldwide, and its incidence is increasing

Treponema pallidum

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Pathogenesis and Immunity

- *T. pallidum* produces no important toxins or enzymes
- *T. pallidum* often infects the endothelium of small blood vessels, causing endarteritis
- tissue destruction and lesions observed in syphilis are primarily the consequence of the patient's immune response to infection
- Immunity to syphilis is incomplete
 - Antibodies to the organism are produced but do not stop the progression of the disease
 - Patients with early syphilis who have been treated can contract syphilis again
 - Patients with late syphilis are relatively resistant to reinfection

Treponema pallidum

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Pathogenesis and clinical features

- *T. pallidum* causes syphilis that has three phases

Primary Syphilis

- spirochetes multiply at the site of inoculation, and a local, non-tender ulcer (chancre) usually forms in 2 to 10 weeks
- The ulcer heals spontaneously, but spirochetes spread widely via the bloodstream (bacteremia) to many organs



Primary syphilis



Primary syphilis

Treponema pallidum

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Secondary Syphilis

- One to 3 months later, the lesions of secondary syphilis may occur
 - maculopapular rash on the palms and soles
 - moist papules on skin and mucous membranes
- Moist lesions on the genitals: **condylomata lata**
- The lesions are rich in spirochetes and are highly infectious, but they also heal spontaneously
- Constitutional symptoms
 - low-grade fever, malaise, anorexia, weight loss, headache, myalgias, and generalized lymphadenopathy



Treponema pallidum

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- About one-third of the early syphilis (primary and secondary) cases will “cure” themselves, without treatment

Latent Syphilis

- occurs in one-third of the early syphilis cases
- no lesions appear, but positive serologic tests indicate continuing infection
- During early stages (1-2 years after 2⁰) the symptoms of 2⁰ syphilis can reappear and patients can infect others
- In the late latent period (can last for many years), no symptoms occur and patients are not infectious

Treponema pallidum

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Tertiary Syphilis

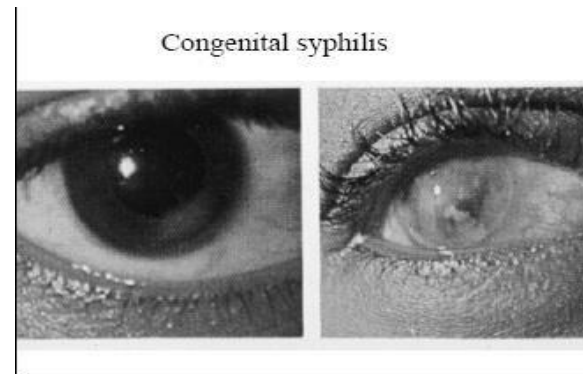
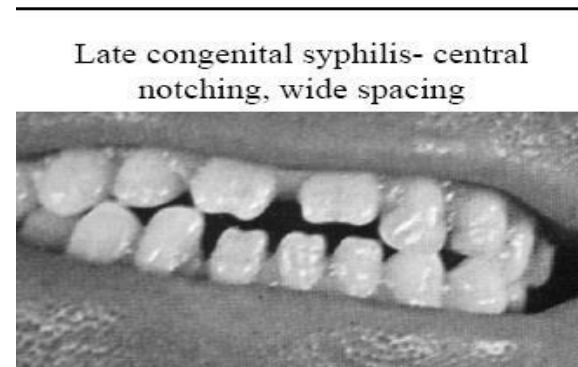
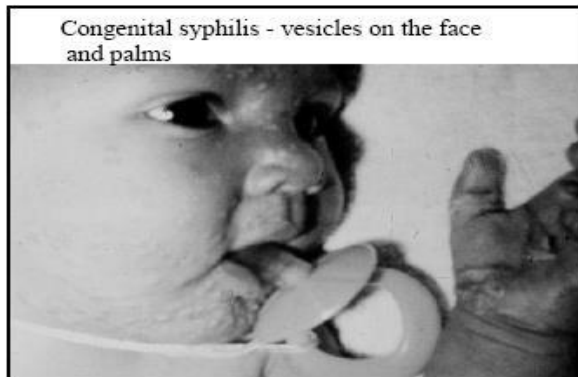
- Occurs in the remaining one-third of early syphilis cases
- develop 3–10 years after the primary infection
- A diffuse chronic inflammation can affect almost any organ or tissue
- **Gummas (granulomatous lesions)** can be found in bones, legs, skin and other tissue
- Lesions in the eyes can lead to blindness
- Lesions in the cardiovascular system can lead to damaged aortic valves
- Neurosyphilis may lead to damage of the spinal cord and paralysis
- In tertiary lesions, treponemes are rarely seen

Treponema pallidum

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Congenital syphilis

- The organism is transmitted across the placenta, typically after the 3rd month of pregnancy (fetal infection)



Treponema pallidum

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Laboratory diagnosis

- Microscopy
 - dark field microscopy
 - Fluorescent antibody test

dark field microscopy



Fluorescent antibody test

Treponema pallidum

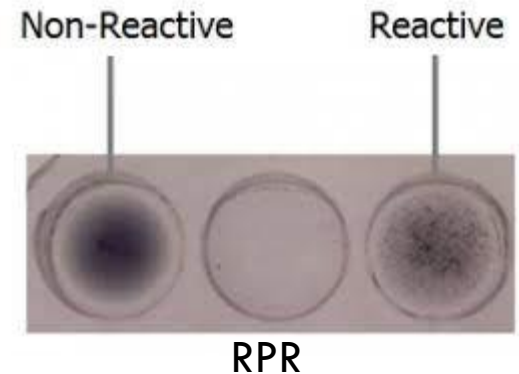
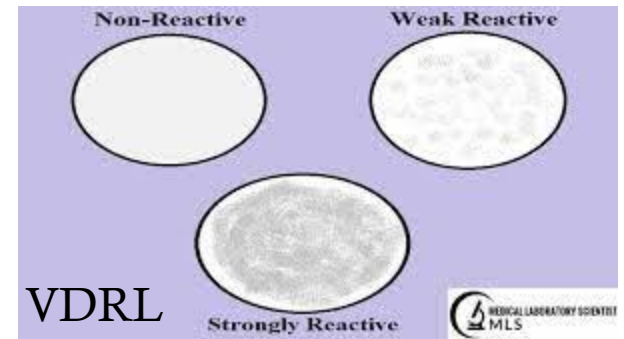
170

Laboratory diagnosis

- Nonspecific Serologic Tests
 - involve the use of nontreponemal antigens
 - Extracts of normal mammalian tissues (e.g., cardiolipin from beef heart)
 - VDRL, RPR
 - False-positive reactions occur
- Specific Serologic Tests
 - involve the use of treponemal antigens and therefore are more specific
 - remain positive for life

Treatment

- Penicillin is drug of choice



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Rickettsia

Rickettsiaceae

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- The family Rickettsiaceae consists of two genera
 - *Rickettsia*
 - *Orientia*
- Genus *Rickettsia* is subdivided into
 - Spotted fever group: *R. rickettsii*, *R. akari*
 - Typhus group: *R. prowazekii* and *R. typhi*
- Genus *Orientia*
 - only a single species: *O. tsutsugamushi*
 - *O. tsutsugamushi* is responsible for the disease scrub typhus

Rickettsiaceae...

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Physiology and Structure

- The Rickettsiaceae are small (0.3×1 to $2 \mu\text{m}$)
- The cell wall structures of Rickettsia are typical of gram-negative rods
 - peptidoglycan layer is minimal (stains poorly with the Gram stain)
 - LPS has only weak endotoxin activity
- Orientia lacks both peptidoglycan layer and LPS
- The Rickettsiaceae are seen best with Giemsa or Gimenez stains

Rickettsiaceae...

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Physiology and Structure

- The Rickettsiaceae are strict intracellular parasites
 - found free in the cytoplasm of infected cells
- They enter host cells by phagocytosis → degrade the phagosome membrane by producing a phospholipase → released into the cytoplasm → Multiply by binary fission (generation time, 9-12 hrs)
- The spotted fever group of Rickettsia and Orientia are continually released from cells through long cytoplasmic projections
- The typhus group accumulates in the cell cytoplasm until the cell membranes lyse, signaling cell death and bacterial release
- The Rickettsiaceae are able to produce ATP, but use host cell ATP as long as it is available

Rickettsiaceae...

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Transmission and Epidemiology

- the Rickettsiae are maintained in nature in certain arthropods such as ticks, lice, fleas, and mites
- are transmitted to humans by the bite of the arthropod
- The rickettsiae circulate widely in the bloodstream (bacteremia), infecting primarily the endothelium of the blood vessel walls
- Rickettsial diseases are zoonotic (i.e., they have an animal reservoir)
 - exception is epidemic typhus, which occurs only in humans

Rickettsiaceae...

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Disease	Organism	Arthropod Vector	Mammalian Reservoir
Spotted fevers			
Rocky Mountain spotted fever	<i>R. rickettsii</i>	Ticks	Dogs, rodents
Rickettsial pox	<i>R. akari</i>	Mites	Mice
Typhus group			
Epidemic	<i>R. prowazekii</i>	Lice	Humans
Endemic	<i>R. typhi</i>	Fleas	Rodents
Scrub	<i>R. tsutsugamushi</i>	Mites	Rodents

Rickettsiaceae...

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Clinical Findings

Typhus

- Epidemic typhus (louse-borne typhus): caused by *R. prowazekii*
- Endemic typhus (flea-borne typhus): caused by *Rickettsia typhi*
- Scrub typhus (chigger-borne typhus): caused by *O. tsutsugamushi*

Rickettsiaceae...

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Laboratory Diagnosis

- Weil-Felix test
- indirect immunofluorescence
- enzyme-linked immunosorbent assay (ELISA)

Treatment

- The treatment of choice for all Rickettsial diseases is doxycycline

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Borrelia species

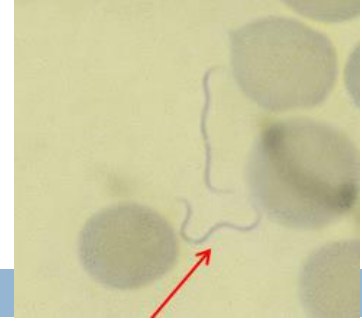
Borrelia

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- Members of the genus *Borrelia* cause two important human diseases
 - **Lyme disease**
 - **Relapsing fever**
- Human lyme disease (Lyme borreliosis)
- is caused by
 - *B. burgdorferi*
 - *B. garinii*
 - *Borrelia afzelii*
- Transmission
 - tick bite

Borrelia...

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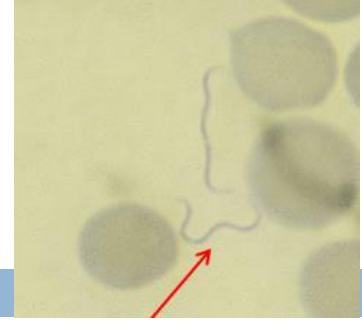


Physiology and structure

- Members of the genus *Borrelia* stain poorly with the Gram stain reagents
 - Considered neither gram positive nor gram negative
 - But have an outer membrane similar to gram-negative bacteria
- Are larger than other spirochetes (0.2 to 0.5×8 to $30 \mu\text{m}$)
- Stain well with aniline dyes (e.g., Giemsa or Wright stain)
- Can be easily seen by light microscopy when present in smears of peripheral blood from patients with relapsing fever

Borrelia...

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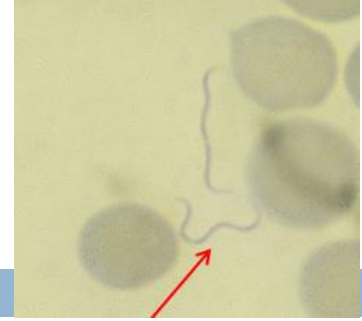


Physiology and structure

- Borreliae are microaerophilic and have complex nutritional needs
- makes them difficult to grow in the lab
- The species that have been successfully cultured have generation times of 18 hours or longer
- Because culture is generally unsuccessful, diagnosis of diseases caused by
 - borreliae is by serology (Lyme disease) or
 - microscopy (relapsing fever)

Borrelia...

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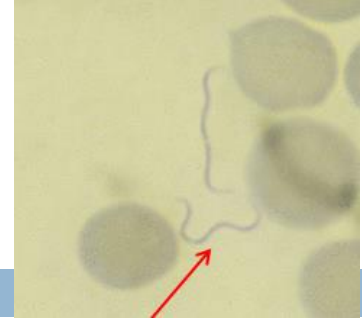


Pathogenesis and immunity

- Borreliae do not produce recognized toxins
- Are removed rapidly when a specific antibody response is mounted
- The periodic febrile and afebrile cycles of RF result from
 - ability of the borreliae to undergo **antigenic variation**

Borrelia...

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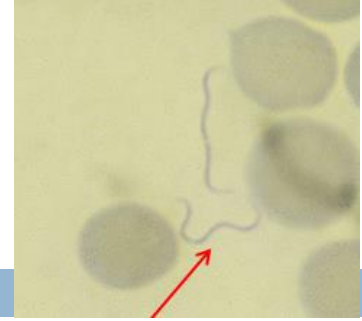
Epidemiology

A. Epidemic relapsing fever

- Caused by *B. recurrentis*
- Transmitted by human body louse
- humans are the only reservoir
- Lice become infected after feeding on an infected person
- Disseminated disease is not believed to occur in lice
- Human infection occurs when the lice are crushed during feeding
- crowded, unsanitary conditions (e.g., wars, natural disasters)
- Found in Ethiopia, Eritrea, Somalia, and Sudan

Borrelia...

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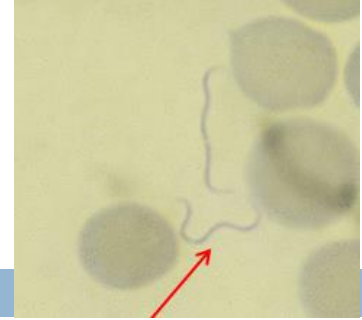
Epidemiology

B. Endemic relapsing fever

- caused by as many as 15 species
- Transmitted by Soft ticks (Ornithodoros species)
- rodents, small mammals, and soft ticks (Ornithodoros species) as the main reservoirs → zoonotic disease
- Unlike the louse-borne infections, the borreliae that cause endemic disease produce a **disseminated infection in ticks**
- The ticks contaminate the bite wound with borreliae present in saliva or feces
- Tick-borne disease is found worldwide, corresponding to the distribution of the Ornithodoros tick

Borrelia...

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Epidemiology

Infection	Reservoir	Vector
Relapsing fever epidemic (louse-borne)	Humans	Body louse 
Relapsing fever endemic (tick-borne)	Rodents, soft ticks	Soft tick 
Lyme disease	Rodents, deer, domestic pets, hard ticks	Hard tick 

Borrelia...

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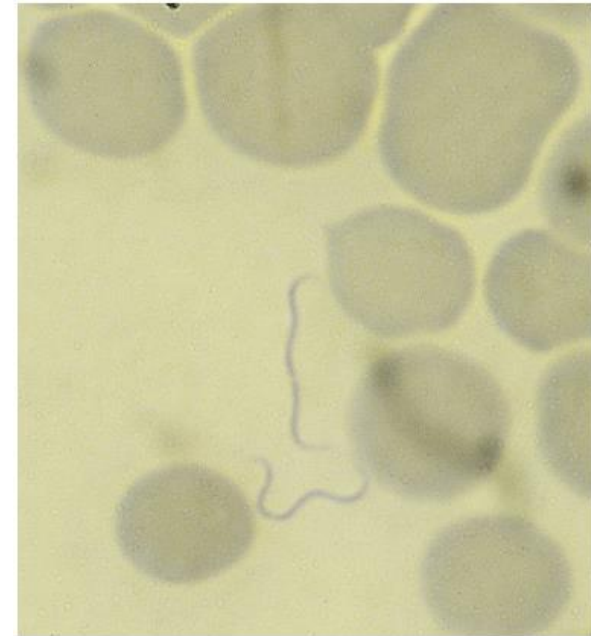
Laboratory diagnosis of RF

A. Microscopy

- Specimen: Blood
- Giemsa-stained or Wright-stained blood film
- most sensitive method for diagnosing RF, with smears positive for organisms in more than 70% of patients

Treatment

- Treated most effectively with **tetracyclines** or **penicillins**
- No vaccine



Borrelia organisms in the blood of a patient with endemic relapsing fever (Giemsa stain)

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Chlamydia trachomatis

Chlamydia trachomatis

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General Characteristics

- Obligate intracellular bacteria (i.e. grow only within cells)
- Have a rigid cell wall without peptidoglycan layer
- Their cell walls resemble those of gram-negative bacteria but lack muramic acid → weakly Gram-negative
- Are ovoid in shape and non-motile
- Are non spore-forming, but the elementary bodies act like spores when released into the host
- Although they synthesize most of their metabolic intermediates, they are unable to make their own ATP and therefore can grow only inside host cells

Chlamydia trachomatis

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- All chlamydiae share a group-specific lipopolysaccharide antigen
- Major outer membrane proteins (MOMP) are species
 - *Chlamydia psittaci* and *C. pneumoniae* each have one immunotype,
 - *C. trachomatis* has at least 15 immunotypes
- Two distinct forms: infectious **elementary bodies** and noninfectious **reticulate bodies**

TABLE 25–1 Chlamydiae of Medical Importance

Species	Disease	Natural Hosts	Mode of Transmission to Humans	Number of Immunologic Types
<i>C. trachomatis</i>	Urethritis, pneumonia, conjunctivitis, lymphogranuloma venereum, trachoma	Humans	Sexual contact; perinatal transmission	More than 15
<i>C. pneumoniae</i>	Atypical pneumonia	Humans	Respiratory droplets	1
<i>C. psittaci</i>	Psittacosis (pneumonia)	Birds	Inhalation of dried bird feces	1

Chlamydia trachomatis

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Elementary bodies (EB)

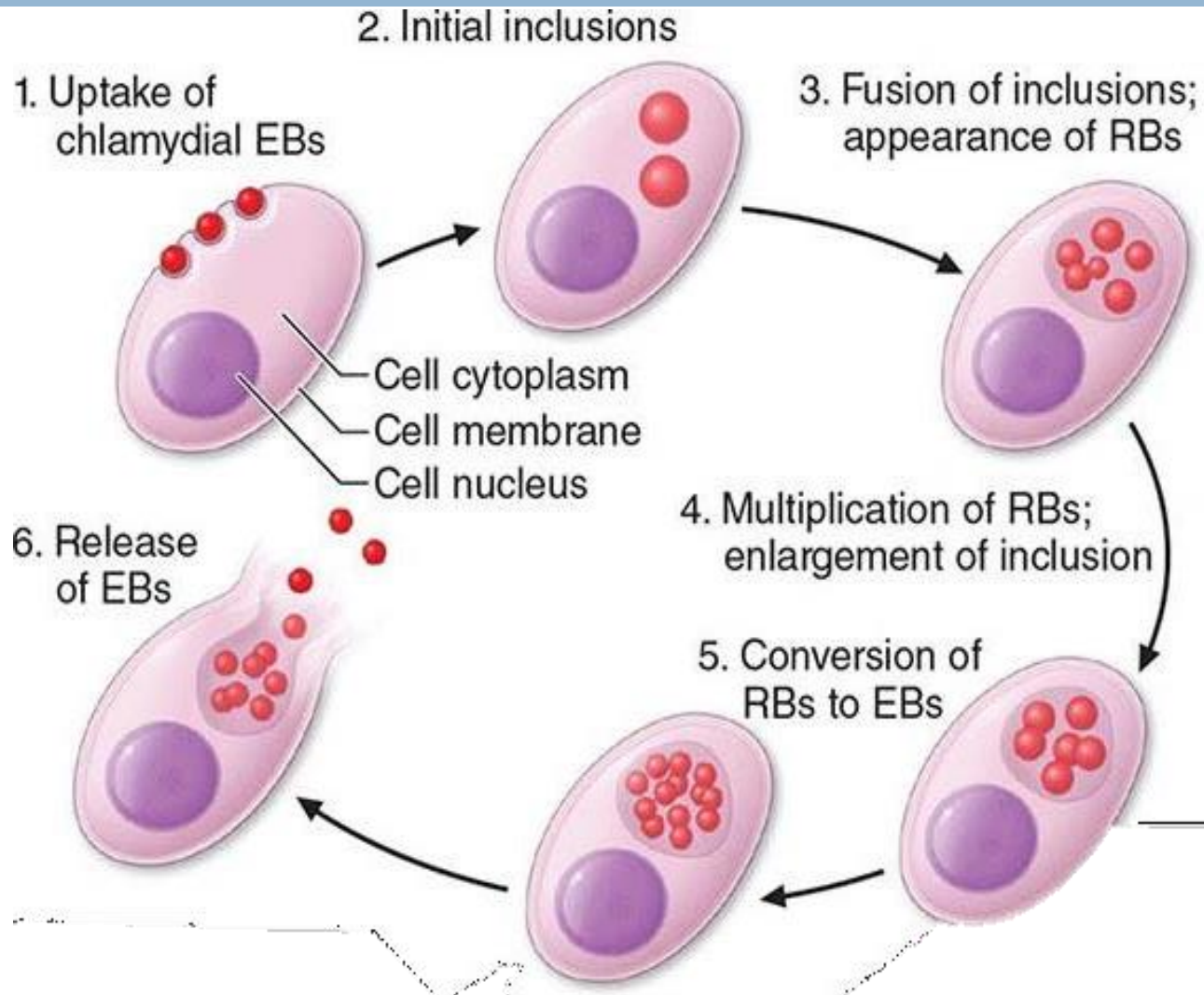
- EBs are small (0.3 - 0.4 μm) infectious form of chlamydia
- possess a rigid outer membrane → resistant to harsh environmental conditions

Reticulate bodies (RB)

- RBs are the non-infectious intracellular form of the chlamydia
- are the metabolically active replicating form of the chlamydia
- possess a fragile membrane

Chlamydiae...Life Cycle

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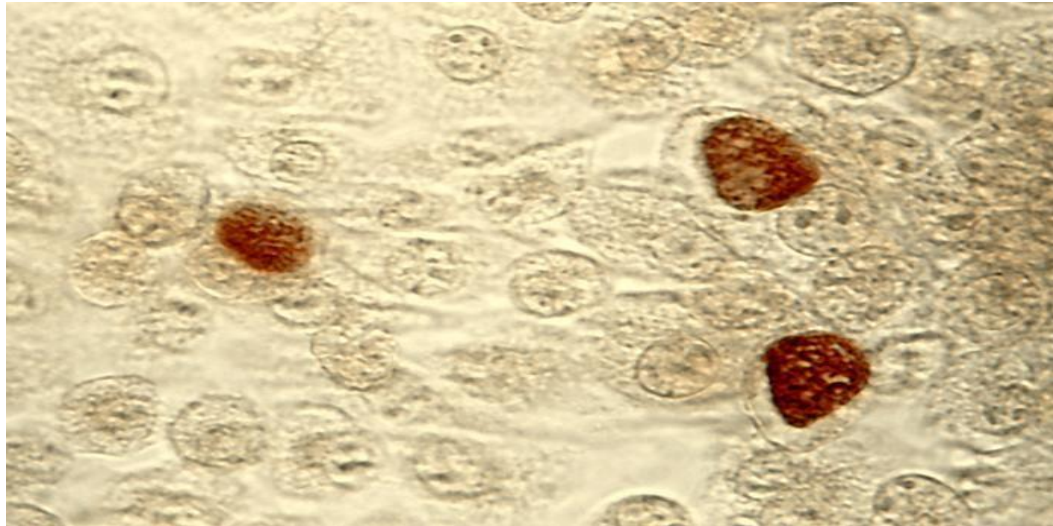
Chlamydiae...Life Cycle...

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1. The EBs bind to receptors on susceptible cells and are internalized by endocytosis and/or by phagocytosis.
2. Within the host cell endosome the EBs reorganize and become RBs.
3. The chlamydia inhibit the fusion of the endosome with the lysosomes and thus resist intracellular killing.
4. The entire intracellular life cycle of the chlamydia occurs within the endosome.
5. RBs replicate by binary fission and reorganize into EBs.
6. The resulting inclusions may contain 100 - 500 progeny.
7. Eventually, the cells and inclusions lyse (*C. psittaci*) or the inclusion is extruded by reverse endocytosis (*C. trachomatis* and *C. pneumoniae*)

Chlamydiae...Life Cycle...

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Chlamydia trachomatis inclusion bodies
(brown) in a McCoy cell culture

Chlamydia trachomatis...

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Transmission & Epidemiology

- infects only humans
- is usually transmitted by close personal contact
 - sexually or by passage through the birth canal
- Individuals with asymptomatic genital tract infections are an important reservoir of infection for others
- In trachoma, *C. trachomatis* is transmitted by finger to-eye or fomite-to-eye contact

Chlamydia trachomatis

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- *C. trachomatis* has a limited host range, with infections restricted to humans.
- The species responsible for human disease are subdivided into two biovars
 1. trachoma
 2. lymphogranuloma venereum (LGV)
- The biovars have been further divided into serovars based on antigenic differences in the MOMP
- Specific serovars are associated with specific diseases



Table 35-3 Clinical Spectrum of *Chlamydia trachomatis* Infections

Serovars	Disease
A, B, Ba, C	Trachoma
D-K	Urogenital tract disease
L1, L2, L2a, L2b, L3	Lymphogranuloma venereum

Chlamydia trachomatis...

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Laboratory Diagnosis

- Specimen: from the involved site (e.g., urethra, cervix, rectum, oropharynx, conjunctiva)
- Antigen Detection
 - Direct immunofluorescence, ELISA
 - neither is as sensitive as culture or nucleic acid–based tests
- PCR: test of choice (90-98% sensitive and very specific)
- Culture: The most specific method (70-85% sensitivity)

Treatment and prevention

- Doxycycline, erythromycin, azithromycin
- No vaccine